Water Market Trends

Updated Trends in Northern Victorian Water Trade 2001-2024



VICTORIA State Government
Energy, Environment and Climate Action

Acknowledgements

DEECA would like to acknowledge the contributions of Frontier Economics for drafting the early versions of this report.

Photo credit

The title page photo was taken by Darryl Whitaker on a farm in Woorinen, Victoria.

We acknowledge and respect Victorian Traditional Owners as the original custodians of Victoria's land and waters, their unique ability to care for Country and deep spiritual connection to it.

We honour Elders past and present whose knowledge and wisdom has ensured the continuation of culture and traditional practices.

DEECA is committed to genuinely partnering with Victorian Traditional Owners and Victoria's Aboriginal community to progress their aspirations.



© The State of Victoria Department of Energy, Environment and Climate Action December 2025.

Creative Commons

This work is licensed under a Creative Commons Attribution 4.0 International licence, visit the ***Creative Commons website*** (***http://creativecommons.org/licenses/by/4.0/***).

You are free to re-use the work under that licence, on the condition that you credit the State of Victoria as author. The licence does not apply to any images, photographs or branding, including the Victorian Coat of Arms, and the Victorian Government and Department logos.

ISBN 978-1-76176-735-7

Disclaimer

This publication may be of assistance to you but the State of Victoria and its employees do not guarantee that the publication is without flaw of any kind or is wholly appropriate for your particular purposes and therefore disclaims all liability for any error, loss or other consequence which may arise from you relying on any information in this publication.

Accessibility

To receive this document in an alternative format, phone the Customer Service Centre on 136 186, email ***customer.service@deeca.vic.gov.au***, or contact National Relay Service on 133 677. Available at ***https://www.waterregister.vic.gov.au/water-trading/trade-reports***.

Contents

Contents 3
1. **About this report** 4
2. **Context** 4
3. **Private water use** 6
4. **Water share trade** 8
5. **Water allocation trade** 11
6. **Water market insights** 19
7. **Conclusions** 24
8. **References** 25

Appendix 1 – Water shares linked to land 26
Appendix 2 – Detailed data on water trade and use 28

1.About this report

The purpose of this report is to provide insights into trends in how water share holders, irrigators and others have been using northern Victorian water markets to access water, manage their water risks and trade water between people and places. It provides an update to the last water market trends report (DELWP 2019) and the original market trends report (DEPI 2016) by focussing on new data and observed trends since 2018-19, in the context of the historic data from previous reports.

Making this information publicly available is part of the Victorian Government's commitment to provide transparency about changes in how water entitlements are held and used in northern Victoria to help water users make informed decisions about their participation in the market.

Unless otherwise stated, the data presented in this report comes from the Victorian Water Register. Data for periods prior to 2007 (i.e. 1991 and 2001) comes from *The Value of Water: A Guide to Water Trading in Victoria* (DNRE 2001). Appendix 2 to this report presents tables and charts of data to allow anyone to quote, use and analyse this information further.

2. Context

Water markets in northern Victoria are a subset of broader markets spanning the southern-connected Murray Darling Basin (including the South Australian Murray, NSW Murray and Murrumbidgee systems). Subject to trading rules, water can move between these interconnected systems (or trading zones).

Water availability and seasonal conditions

The last water market trends report (DELWP 2019) was released during a period of low rainfall and inflows across the southern Murray Darling Basin. Water availability was particularly low in NSW, where general security allocations were 0% and 7% in the NSW Murray and Murrumbidgee respectively.

Dry conditions continued through 2019-20, however water availability began to improve in 2020-21, which was then followed by three relatively wet years with high water availability for entitlement holders and widespread flooding in northern Victoria. The relatively wet conditions challenged river operators and irrigators with full and spilling storages and waterlogged land. The Resource Manager announced 100% seasonal determinations to low-reliability water shares in the Victorian Murray for the first time ever in 2021-22, which was then repeated in 2022-23 and 2023-24.

Regional commodity trends and water use behaviour

The area of perennial horticulture plantings along the Murray downstream of the Barmah Narrows in NSW, South Australia and Victoria has continued to grow, increasing by 23,000 hectares between 2015 and 2021. This increase has been most notable for table grapes and nuts, with some crop plantings such as wine grapes declining over this period (SunRISE 2022). Expansion of the irrigated area of table grapes and nuts in the region continued between 2021 and 2023 (SunRISE 2023). This expansion reflects favourable commodity prices and proven farming systems for profitable growing of these crops in the region. It is in this context that these horticulturalists can often 'out compete' other water market participants when water is scarce by offering higher prices for water. The trends data in sections 3 and 5Figure 1 of this report highlights that private water users in the Victorian Murray downstream of Barmah Narrows (Lower Murray Water (LMW) diverters and LMW districts), have retained strong and consistent water use with the support of large volumes of trade. Since 2010, there has also been a shift in seasonal cropping patterns in the Murrumbidgee system driven by commodity prices and the development of colder climate varieties of cotton where large areas of irrigated cotton have replaced more traditional rice crops (BoM 2023). While areas of irrigated cotton fell in 2019-20 with low water availability and corresponding high water prices alongside softening cotton prices, areas of both irrigated cotton and rice have since increased with improved water availability (BoM 2023).

Water recovery for the environment

Water recovery targets from the largest irrigation modernisation project in Australian history, the Goulburn-Murray Water (GMW) Connections Project, were finalised in 2020. Consistent with commitments under the project, this resulted in reductions in GMW's loss allowance entitlement as a result of water savings from improved efficiency, and the issue of new water entitlements to environmental water holders, Melbourne's urban water retailers, and Goulburn Murray Irrigation District (GMID) delivery share holders.

The purchase of water shares for the environment in northern Victoria under the Commonwealth Government's Restoring the Balance in the Murray-Darling Basin Program, which peaked in 2009-10 to 2011-12, was wrapped up by 2018. Between 2018-19 and 2023-24 there were no water share purchases for the environment in northern Victoria. However, in 2023 Commonwealth Government announced plans to purchase additional water towards Basin Plan water recovery, and under the Aboriginal Water Entitlements Program, across the Murray-Darling Basin, including in northern Victoria. These programs commenced in 2024-25 and are expected to continue into 2025-26 and 2026-27. Volumes of water shares purchased by the Australian Government after 1 July 2024 can be viewed ***here***.

Water market and Victorian trade rule reform

Over the period 2018 to 2022, DEECA completed a review of the Goulburn to Murray trade arrangements. The review was established because unprecedented volumes of traded water delivered from the Goulburn to the Murray in 2017-18 and 2018-19 caused environmental damage to the lower Goulburn River with sustained high flows in summer and autumn. The review process resulted in the introduction of restrictions on use from tagged accounts when allocation trade limits apply (introduced initially in December 2019) and new Goulburn to Murray trade and operating rules (introduced initially on 1 July 2021). Details of the changes made to the rule can be found ***Goulburn to Murray trade review Long-term rules – fact sheet 1***.

In response to community concerns about unethical behaviour in water markets, the Australian Competition and Consumer Commission (ACCC) was directed in 2019 to conduct an inquiry into Murray-Darling Basin water markets, including their operation, transparency, regulation, competitiveness and efficiency. This Inquiry was completed in 2021. The ACCC did not find evidence that investors exercised market power or manipulated markets to increase water prices in the Southern Connected Basin. However, the ACCC did find that the perception of misconduct by water investors has reduced water users' confidence in water markets and that information was not readily available to robustly assess behaviour and market performance. The ACCC made several recommendations related to water market transparency and integrity which Victoria and the other Basin governments have used to develop a roadmap for water market reform.

Two key Victorian market transparency improvements relevant to this report have been made in recent years. Firstly, in August 2020, Victoria introduced a new requirement that the 'reason for trade' be recorded on all online allocation trade applications (NSW and South Australia have both also introduced similar requirements). This change has enabled improved reporting and market information about why people are using allocation trade. The 'reasons for trade' categories are: standard commercial trade, related party trade, carryover parking, carryover parking return, private lease arrangement, allocation trade as part of a water share transfer, forward contract and 'other'. Secondly, in response to community consultation, changes to Victoria's *Water Act 1989* were introduced in July 2024 that enable new information to be published about frequent allocation traders (people who make more than 20 commercial Murray and/or Goulburn allocation trades in a year).

Water delivery in the Murray River

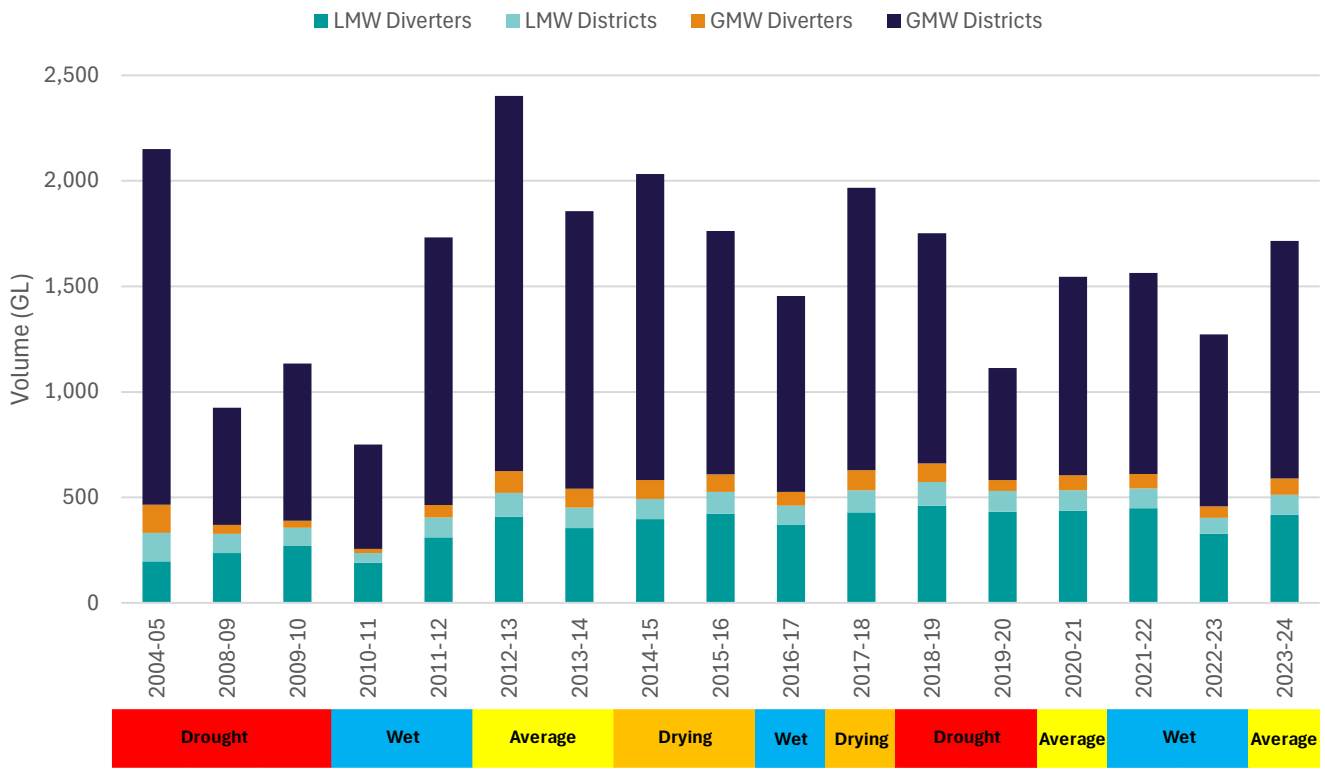
Another issue, indirectly related to water markets, is increasing delivery or shortfall risks in the Murray. There have been significant changes across the connected Murray-Darling Basin over the last 10-20 years that have made it increasingly difficult to deliver water in the Murray downstream of the Barmah Narrows. These changes include the reduced channel capacity through the Barmah Narrows, less frequent access to water in the Menindee Lakes and changing timing and location of water demand.

In addition to working with the other Murray-Darling Basin governments to understand, manage and prepare for shortfalls, Victoria introduced in November 2023 a new 'place of take approvals' framework to clarify and protect the rights of water users to have water delivered down rivers in a shortfall and enable more flexibility for river diverters to manage their own delivery risks through the trade of extraction share. The newly introduced extraction share market is not covered in this report, due to the limited data available to date, however future water market reporting will expand to incorporate extraction share trade information.

3. Private water use

Where private parties (i.e. not water corporations or environmental water holders) have been using water in northern Victoria, and how this has changed over time provides useful context for understanding activity and trends in the water market. Figure 1 presents private annual water use in northern Victoria by GMW and LMW river diverter (diverter) and irrigation district (district) customers against a backdrop of different seasonal conditions (drought, drying, average and wet).

Figure 1: Annual private water use in northern Victoria



Notes:

1. *Seasonal indicators are based on northern Victoria rainfall records, and not water availability. They are indicative only for the purpose of highlighting when irrigation areas were wetter or drier in the lead up to and during the irrigation season.*
2. *Water use data from 2004-05 has been sourced from water corporation annual reports.*

Since 2018-19, private annual water use has fluctuated between 1,112 GL and 1,751 GL. Water use was lowest in 2019-20 when dry conditions resulted in low water availability across the southern connected Murray-Darling Basin, although notably, use was higher than in the last drought year in 2008-09. Low-reliability allocations were available in the Murray in 2021-22, 2022-23 and 2023-24, but there is no observable consequent increase in water use in those years in this data. There is also no obvious signal of decreased private water use from the acquisition of water shares for the environment between 2009-10 to 2018; it is likely that year-to-year climate and water availability variations cause sufficient 'noise' in the data to mask such a trend if it exists.

GMW district customers have continued to account for the largest share of private water use in northern Victoria, however their use volumes are also the most variable. The decline in water use in 2019-20 was most evident in GMW's districts where only 531 GL was used (almost half their average annual use).

LMW diverters and district users had a comparatively small decline in water use in response to the dry conditions. The difference in GMW and LMW regional experiences highlights the stark difference between highly variable water use in GMW's region (where irrigation is dominated by mixed farming and dairy) and the more consistent water use in LMW's region (where irrigation is dominated by perennial horticulture).[1]

Figure 1 shows that with the exception of 2022-23 when widespread flooding suppressed demand, use by LMW diverters has been relatively stable since 2017-18, indicating that growth in perennial horticulture plantings during this time has not increased overall water use. This contrasts with the trend observed in the 2018 Market Trends Report that water use by LMW diverters had increased since 2004-05. It is likely that the stable trend since 2017-2018 – even during the 'drought' years of 18-19 and 19-20 – is influenced by several factors, including the relatively mild seasonal conditions since 2020-21, exits of existing farm businesses, and reductions in local seasonal cropping.

[1] GMW's irrigation districts are mostly dominated by cropping and dairying but do include important areas of non-dairy grazing and horticultural plantings (GBCMA 2023), and GMW's river diverters (irrigators who pump their own water directly from rivers without being part of an irrigation district) irrigate a range of enterprises, predominantly cropping, non-dairy grazing and horticulture (GBCMA 2024). LMW's irrigation districts comprise predominantly permanent plantings of grapevines, and LMW's river diverters overwhelmingly grow perennial horticultural crops. Almonds are now the dominant crop, but there are also extensive plantings of grapevines, citrus, and various other crops (MCMA 2022).

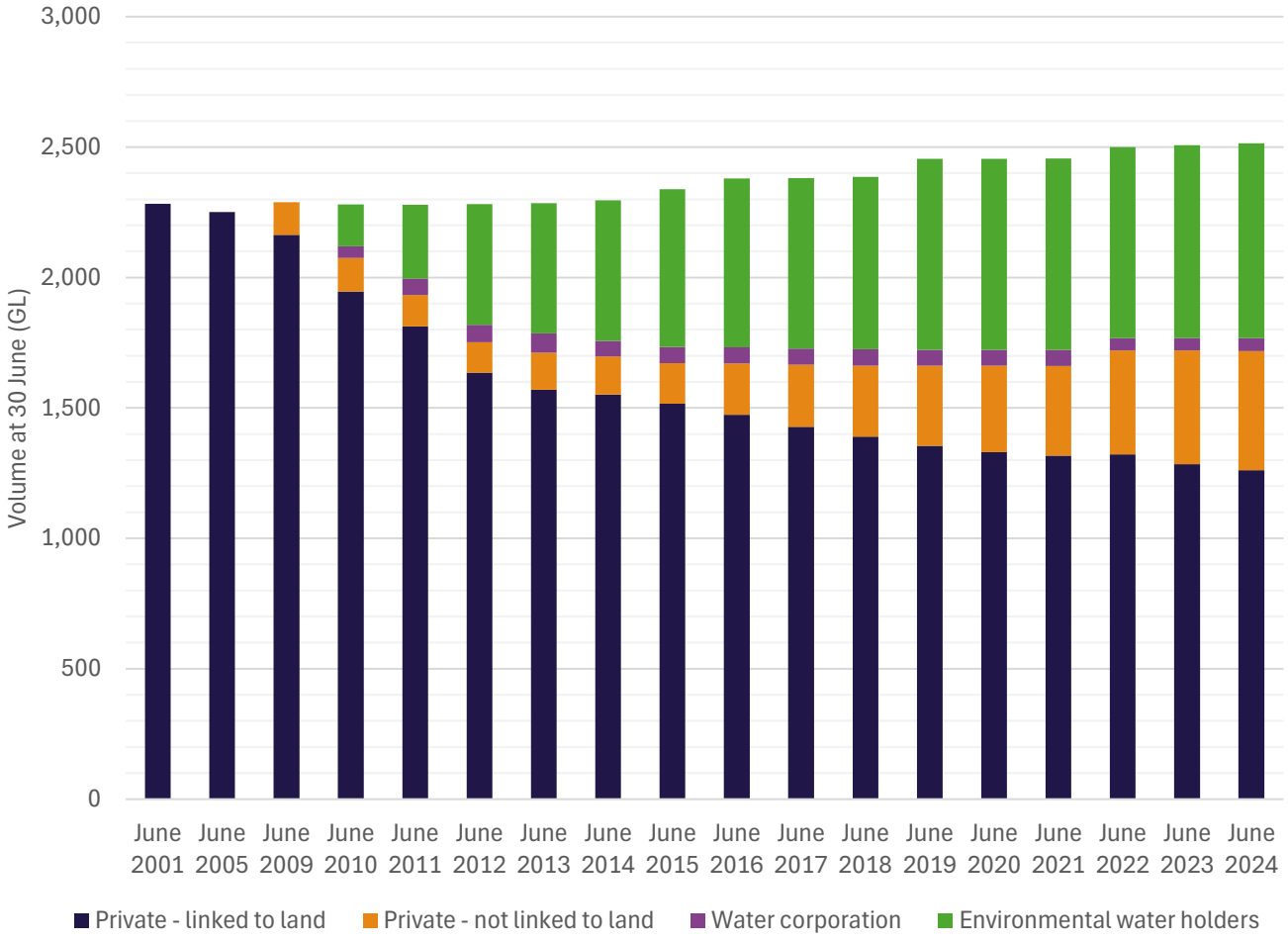
4. Water share trade

Water share ownership

Trends in water share ownership highlight who has access to water allocations that could be made available on the market, and how that has changed over time. Owners can choose to link their water shares to land which may indicate their intent to use the water allocations on that land, although does not prevent them selling allocations on the market. Owners may choose not to link their water shares to land for a wide variety of reasons, including financial and accounting purposes[2], irrigators wanting flexibility to move water for use on their land and investors selling water allocations to generate an annual return.

Figure 2 presents the ownership and nature of holding high-reliability water shares at 30 June each year. The method of reporting on "links to land" in this report differs from previous reports in response to changes to the Water Act 1989 that were made in 2023 (described in detail in Appendix 1 – Water shares linked to land).

Figure 2: Changing ownership of high-reliability water shares in northern Victoria



[2] For example, some may hold water shares in their superannuation account, so allocations are transferred to the required farm allocation account (rather than linked) for use.

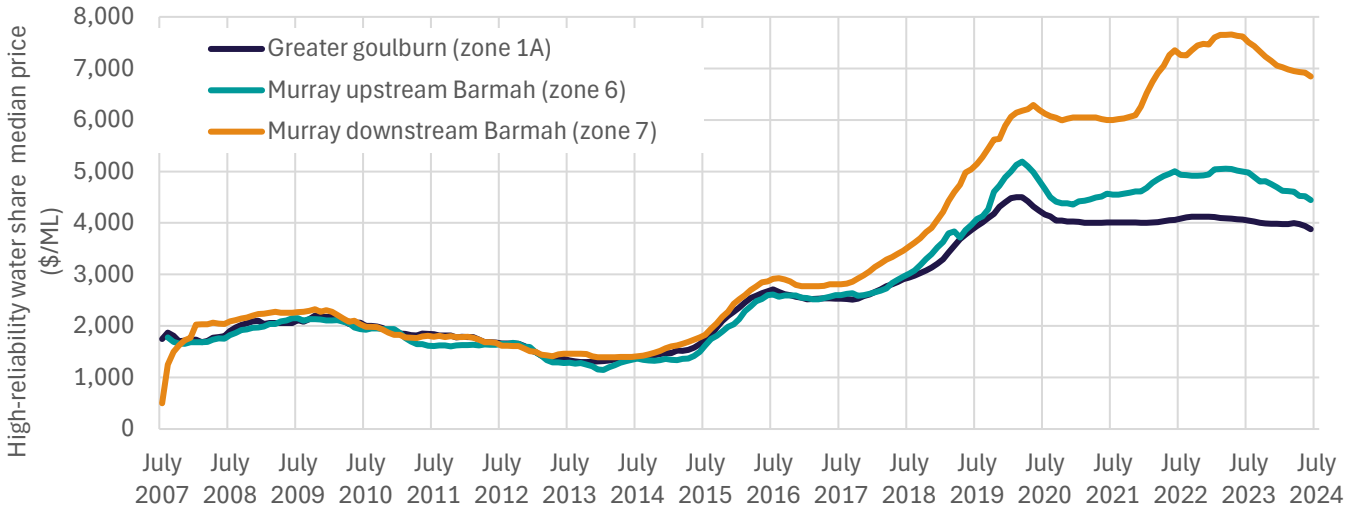
Since 2018, there has been no significant change in the proportion of high-reliability water share volumes in northern Victoria held by the environment (30%), water corporations (2%) and private owners (68%). However, the total volume of high-reliability water shares in northern Victoria has increased by 129 GL during this period as a result of entitlement changes reflecting water recovered under the GMW Connections Project.[3]

Most privately held high-reliability water shares remain linked to land, however this volume has continued to decline, reducing from 1,390 GL at 30 June 2018 to 1,262 GL at 30 June 2024. The rate of decline has been higher in irrigation districts than for diverters across GMW and LMW regions[4]. At 30 June 2024, 18% of high-reliability water shares in northern Victoria were privately owned and not linked to land compared to 11% at 30 June 2018. Use of allocations made to water shares not linked to land is explored further in Section 6 of this report.

High reliability water share prices

Prices paid for water shares indicate the value that water users and investors have placed on those entitlements. Figure 3 shows how the median prices of high-reliability water shares in the three largest northern Victorian source trading zones has varied since unbundling in July 2007.

Figure 3: High-reliability water share median prices (6-month rolling average)



Unlike the preceding decade, the Commonwealth was not a purchaser of Victorian water shares between 2018 to 2024[5], with most trade occurring between private parties.

High-reliability water share prices approximately doubled between 2018 and 2020, reflecting the premium value that water users placed on these entitlements during drought conditions. In late 2020 prices fell slightly, but then Murray downstream from Barmah high reliability water share prices subsequently increased again, peaking in early 2023.

[3] This included the issue of 84 GL for the environment and 60 GL to GMID delivery share holders (commonly referred to as the irrigator's share of water recovery) and the surrender of 15 GL held by GMW on behalf of the GMW Connections Project. Under the irrigators' share GMID irrigators were issued entitlement equivalent to 77 GL of long-term average annual yield, which was made up of both high-reliability and low-reliability water shares.

[4] Compared to 2017-18, volumes linked to land in 2023-24 have declined by 10% for GMW districts, 6% for GMW diverters, 15% for LMW districts, and 5% for LMW diverters. Refer to Table 6 in Appendix 2 for volumes.

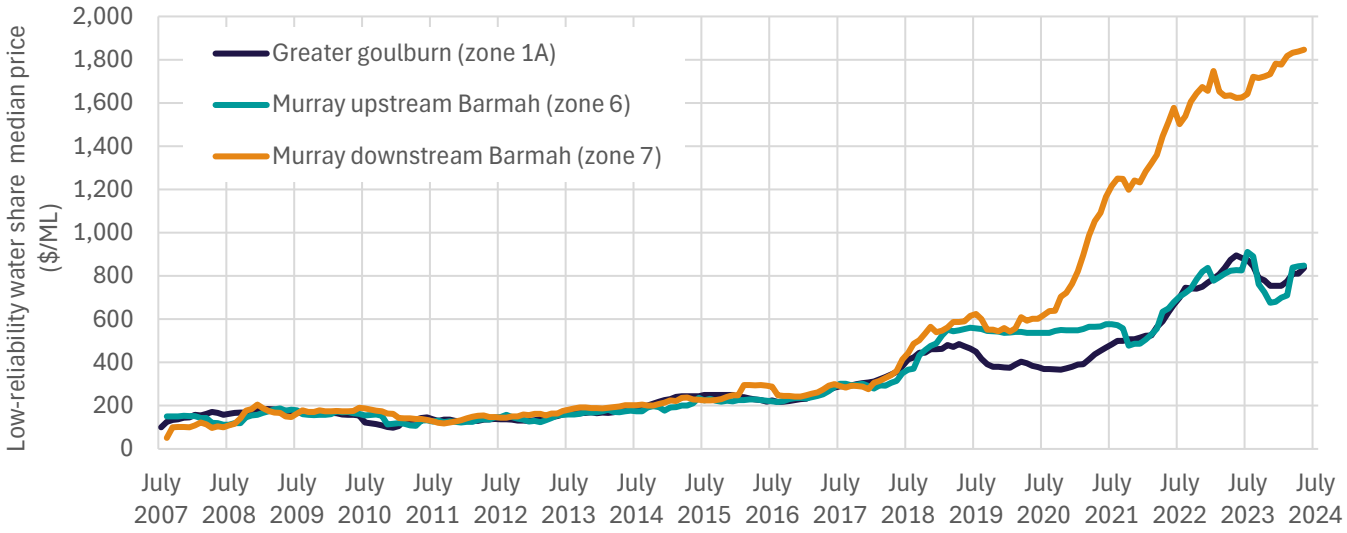
[5] Approximately 1 GL of high-reliability water share was transferred to the Commonwealth between 2019 and 2021, however these were not market purchases, they were held by Victorian agencies and relate to projects recovering additional water for the environment. Since 2024, the Commonwealth has again become a purchaser of Victorian water shares. Information about this can be found on the Victorian Water Register website ***here***.

Water share prices have historically been relatively similar across different water sources. However, since 2018 prices have been noticeably higher in the Murray zone 7 (downstream of Barmah) reflecting growth in perennial horticulture demand downstream of Barmah (including in NSW and South Australia) with its high willingness to pay for water and more regular periods of restriction on allocation trade into the region (explored further in Section 5).

Low reliability water share prices

Figure 4 shows how the median prices of low-reliability water shares has varied since unbundling in July 2007.

Figure 4: Low-reliability water share median prices (6-month rolling average)



In this period, low-reliability water share median prices have followed a similar pattern to high-reliability prices – broadly stable until approximately 2018-19, followed by a series of significant increases. This price increase has been most significant in Murray zone 7, where low-reliability water share prices increased from $444 per ML in July 2018 to $1,861 per ML in June 2024. Low reliability price rises may be influenced by a growing recognition of the value of these water shares to carry over water in average or dry years, as well as receiving allocations in wetter years like 2021-22, 2022-23 and 2023-24.

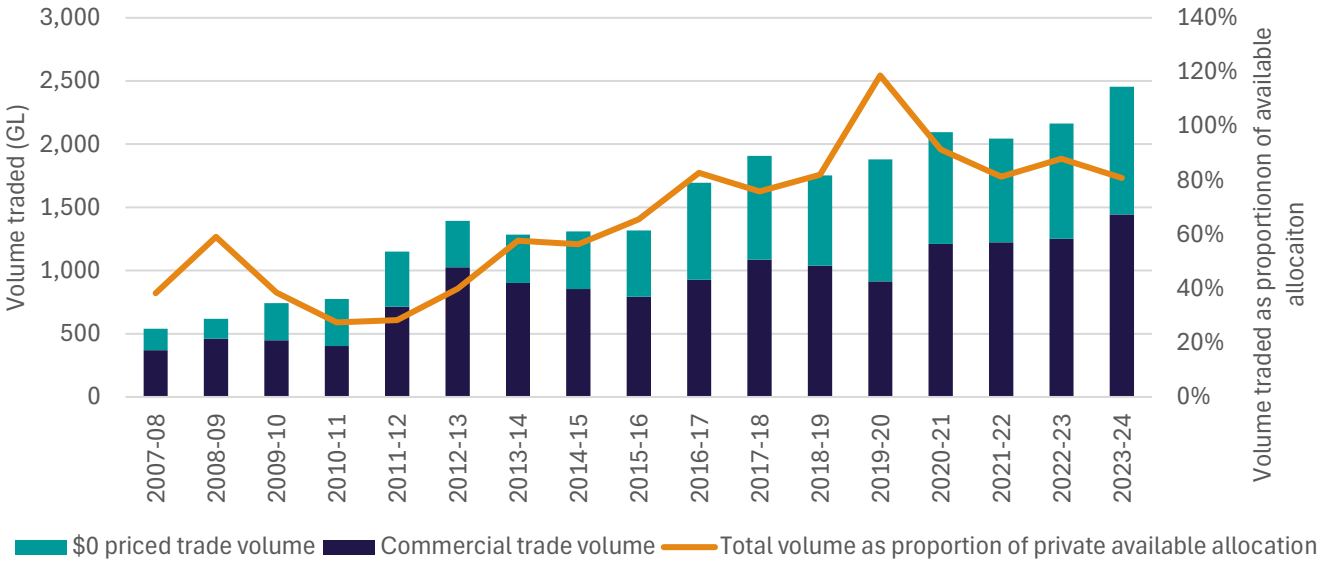
Water share prices in recent years have likely also been supported by emergence of the “carryover parking” market where water share holders have been able to generate new revenue by leasing water shares (representing additional carryover capacity) to others seeking to increase the ability to hold on to their unused allocations. This new market is explored in further detail later in section 5.

5. Water allocation trade

Allocation trade volumes

Annual allocation trade volumes[6] have generally increased year on year since unbundling occurred in 2007-08; the exceptions to this occurred in some of the drier years with low early season allocations (e.g. 2013-14 and 2018-19) (Figure 5). Over this period, trade volumes as a proportion of total allocation available (including volumes carried over and new season allocations) has been trending upwards, indicating ongoing growth of water markets as water users increasingly use allocation markets to maximise the value of their entitlements and meet their water needs within the water year.

Figure 5: Summary of annual allocation trade in northern Victoria



Notes:

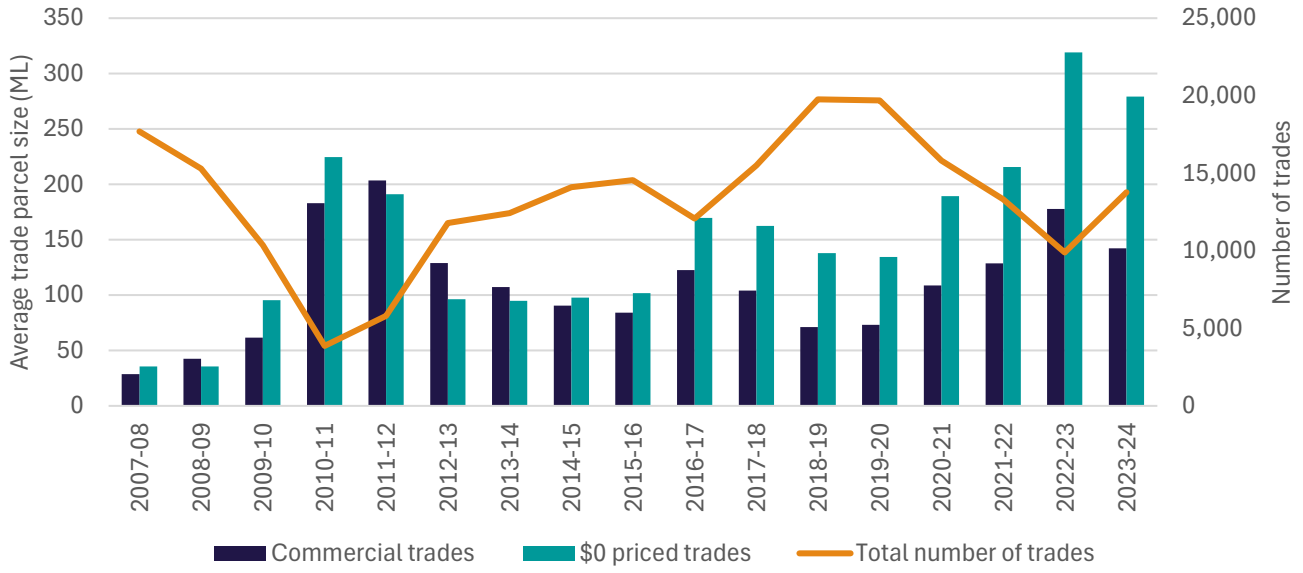
1. *Excludes trade of environmental water. Environmental trade is defined as trades between environmental accounts, or $0 priced trades involving an environmental account.*

Number of allocation trades

The number of allocation trades occurring in drier years has tended to be higher than in wetter years, reflecting that when conditions are drier, market participants tend to trade water in smaller parcels (Figure 6).

[6] The allocation trade analysis in this report excludes $0 trades between environmental water holders because these trades reflect environmental water holders moving water between their accounts or delivering water to another state, so do not contribute to the water available in the market or for private water use. The analysis in this report does, however, include information about allocation trades between the environmental water holders and others (i.e. private entities and water corporations). Further information about environmental water delivery and trade between environmental water accounts is included in the ***Victorian Water Accounts | DEECA Victorian Water Accounts***.

Figure 6. Number of trades and average trade parcel size in northern Victoria



Notes:

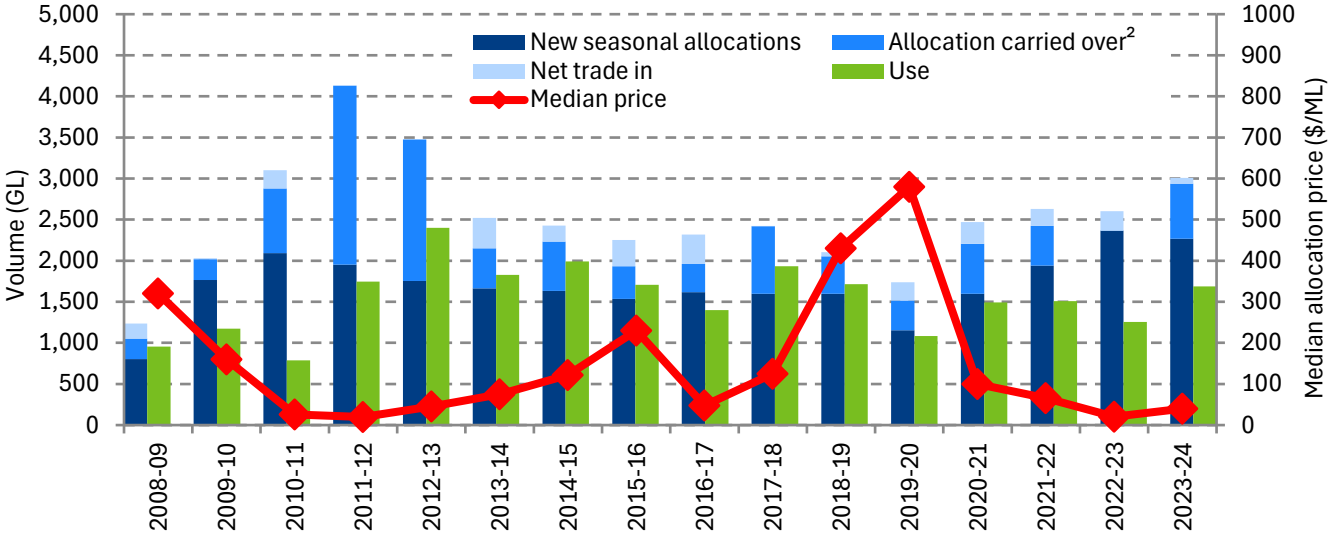
1. *Excludes trade of environmental water. Environmental trade is defined as trades between environmental accounts, or $0 priced trades involving an environmental account.*

Allocation prices in the Goulburn and Victorian Murray

Allocation price trends demonstrate the willingness of market participants to pay for water allocations, and how this has varied over time. Several supply and demand factors influence allocation prices including current and forecast water availability, climate conditions, commodity markets, restrictions on trade and demand. The Goulburn and Victorian Murray are the two largest water supply systems in northern Victoria and have the deepest, most active markets, providing the most robust overall indication of willingness to pay for water allocations in northern Victoria.

Figure 7 presents the trend in annual median allocation prices for the Goulburn and Victorian Murray systems, and how this compares to private water availability and use each year

Figure 7: Victorian Murray and Goulburn private water availability, use and median allocation price



Notes:

1. *Only privately held accounts with a Goulburn or Murray source trading zone are included in this data.*

2. *'Allocation carried over' reflects the carried over water available after deductions from spillable water accounts.*
3. *In every year except for 2012-13, there was a net trade in to Murray and Goulburn private accounts (i.e. more water traded in than out). In 2012-13 there was a net trade out of 1 GL in 2012-13. Since this graph only shows privately held Goulburn and Murray accounts, 'net trade out' may mean trade to interstate or other Victorian systems, or trade to environmental water holders or water corporations within Victoria.*
4. *There was no available allocation carried over in 2022-23 as 100% of carryover water spilled in accordance with the carryover rules.*

Figure 7 shows that allocation prices have continued to be highest during drought and low water availability periods, and lower during wet years with high water availability and low water use.

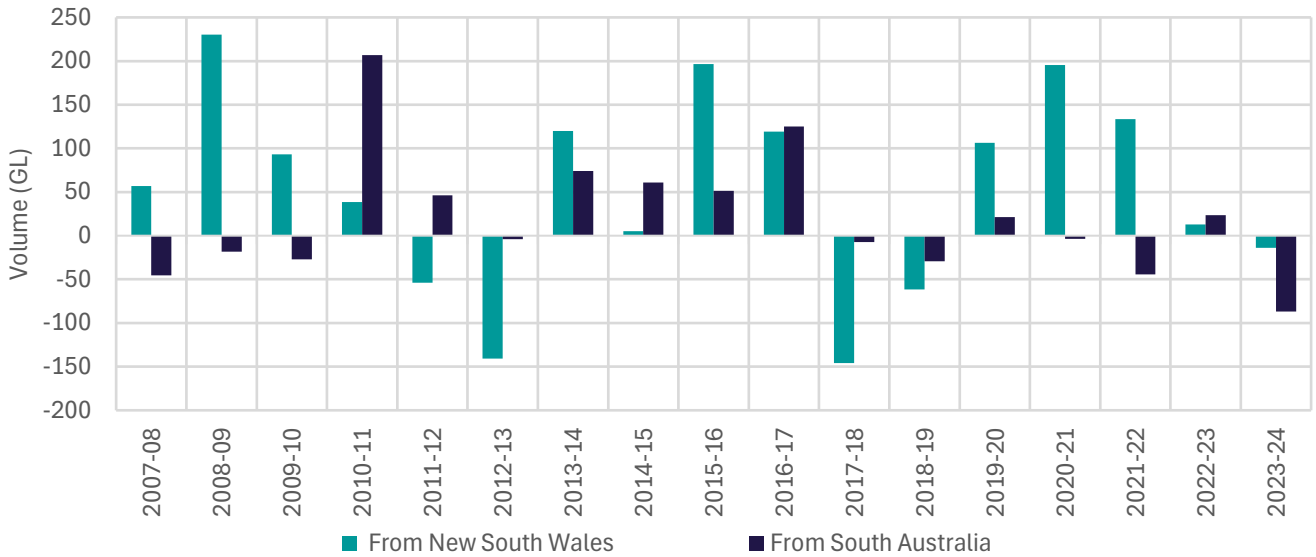
Record high allocation prices were observed in 2019-20 when water availability fell to the lowest levels since the Millennium Drought. Several factors contributed to these higher prices, including two years of hot, dry conditions, very low water availability in NSW (much lower than in 2008-09), the reduced volume of high-reliability water shares held for consumptive purposes (due to purchases for the environment), and growth in perennial horticultural industries with a high willingness to pay for allocation.

The relatively low allocation prices observed since 2020-21 reflect high water availability (high carryover volumes and 100% allocations, including for low-reliability water shares). In 2022-23, allocation prices were particularly low (the median annual price in the Goulburn and Victorian Murray was $20/ML) due to suppressed demand following heavy rainfall and flooding in spring and early summer that offset irrigation needs and damaged crops in some regions. The last time allocation prices have been as low as 2022-23 was in 2011-12, when water availability (new season allocation plus carryover) was even higher.

Interstate allocation trade

Trends in allocation trade into and out of northern Victoria with other parts of the southern connected Basin highlight the differing supply and demand conditions in each of the states over time. Figure 8 presents the net allocation trade into Victoria from NSW and South Australia each year.

Figure 8: Net trade of allocation into Victoria (excluding trade of environmental water)



Notes:

1. *Net trade in this figure includes interstate tagged usage.*
2. *Excludes trade of environmental water. Environmental trade is defined as trades between environmental accounts, or $0 priced trades involving an environmental account.*
3. *Positive values in this figure indicate water moving into Victoria. Negative values indicate water moving out of Victoria.*

Figure 8 shows that in most years, Victoria continues to be a net importer of allocation from the other states. This trend was reversed in 2017-18 and 2018-19 when NSW general security allocations fell below 100% and demand from seasonal crops such as cotton remained high (BoM 2023). While NSW general security allocations remained

low in 2019-20 (3% in the Murray and 11% in the Murrumbidgee), Victorian high-reliability allocations also fell below 100% and opportunities to trade water out of the Goulburn were lower than the previous years, resulting in Victoria again becoming a net importer of allocation supported by the willingness of perennial horticulture (the largest proportion of which is located in Victoria) to pay high allocation prices.

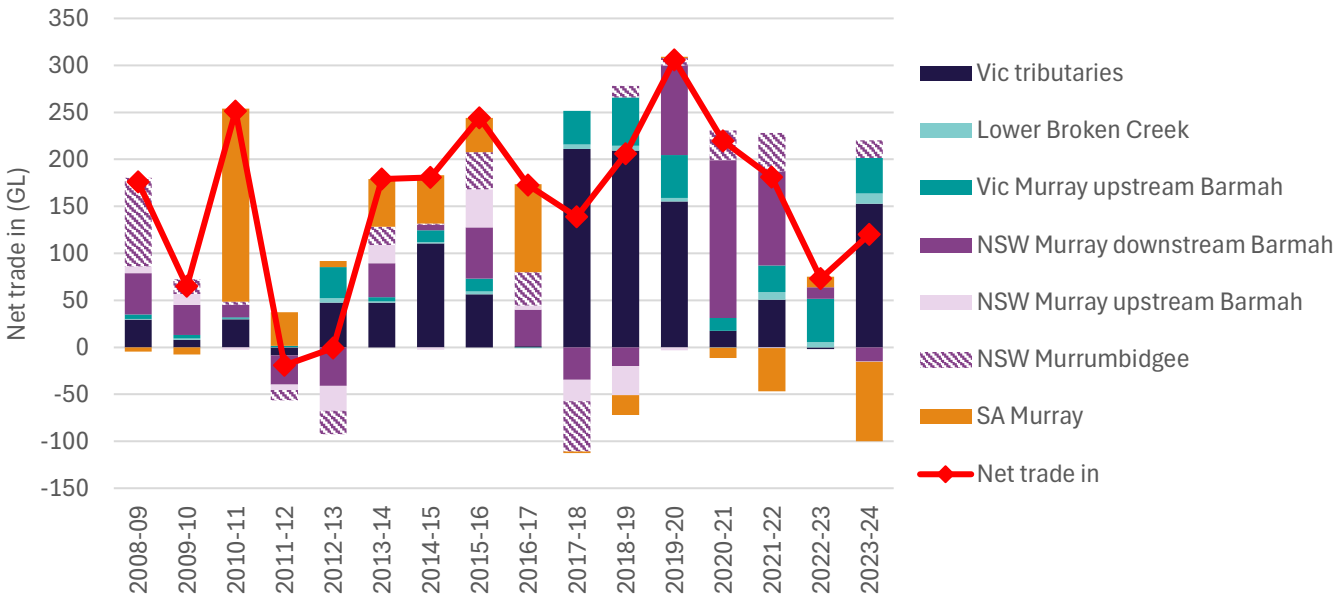
There was relatively little trade between Victoria and NSW during 2022-23 and 2023-24 when high spill risks in the Murray led to restrictions on trade between the two states applying for large periods of these years[7] [8].

Victoria's volume of trade with South Australia has typically been much lower than trade with NSW. Nevertheless, Figure 8 highlights an increasing trend of net trade from Victoria to South Australia – as observed in five out of the last seven years, including 87 GL of net private allocation trade into South Australia in 2023-24. This trend is likely to be influenced by several factors which may include growth in irrigation demand from high value crops in South Australia and lower water availability in NSW during dry periods.

Regional allocation trade – Murray zone 7 (downstream of Barmah)

Net allocation trade into the Murray zone 7 (downstream of Barmah) is worthy of particular consideration given the historically strong demand for water in this reach, and the more recent growth in perennial horticulture plantings downstream of Barmah. Figure 9 shows the net volume of allocation imported into Murray zone 7 by traders, and where that trade was sourced from, including whether it was traded in from interstate, upstream of Barmah and tributaries.

Figure 9: Net allocation trade (including tagged use) into Murray zone 7



Notes:

1. *Victorian tributaries include trading zones 1A, 1B, 2, 3, 4A, 4C, and 5A.*
2. *NSW Murray upstream Barmah includes trading zones 10 and 10B. NSW Murray downstream Barmah includes trading zones 11 and 14.*
3. *This chart excludes trade of environmental water. Environmental trade is defined as trades between environmental accounts, or $0 priced trades involving an environmental account.*

[7] Allocation trade from New South Wales to Victoria is limited to the lesser of a net annual volume of 200 gigalitres (GL) or a volume that keeps the risk of spill in Victoria's share the Murray system below 50 per cent. The spill risk limit currently allows more than 200 GL of trade, and so the net annual volume of 200 GL applies unless advised otherwise. See ***Understanding the NSW to Victoria trade limit - Water Register***

[8] Allocation trade into NSW was restricted for most of 2022-23, and until September of 2023-24, to protect NSW Murray entitlements from an increased spill risk due to trade in. *Temporary restriction interstate water trades into NSW Murray, Lower Darling and Murrumbidgee regulated water sources | NSW Government Water*

Figure 9 shows that Murray zone 7 has historically been a net importer of allocation, with traders importing between net 73 GL and net 306 GL allocation each year since 2017-18. This demonstrates the ongoing demand for water in the zone, which peaked during the recent drought conditions.

Victorian tributaries have been the most consistent supplier of allocation to the region during this period. The contribution of the Victorian tributaries was most significant in 2017-18 and 2018-19 when record volumes of allocation were traded out of the Goulburn system under the now superseded trade rule.[9] The NSW Murray downstream of Barmah became the dominant source of supply in years when NSW had full general security allocations and trade into Victoria was unrestricted for most of the year (i.e. 2020-21 and 2021-22). Prior to 2017-18, South Australia was a major supplier of allocation to the region, although in more recent years this trend has reversed, for example in 2023-24 private traders in zone 7 exported a net 85 GL to South Australia. Where zone 7 private buyers source their water from is likely influenced by a range of factors including water availability, restrictions on trade, demand for water and willingness to pay.

Interzone allocation trade restrictions and price differentials

Trade limits are necessary to protect entitlement holders and waterway health from potential negative impacts. When these limits are reached and allocation trade between zones is restricted, allocation prices in the different zones can diverge in response to different supply and demand conditions.

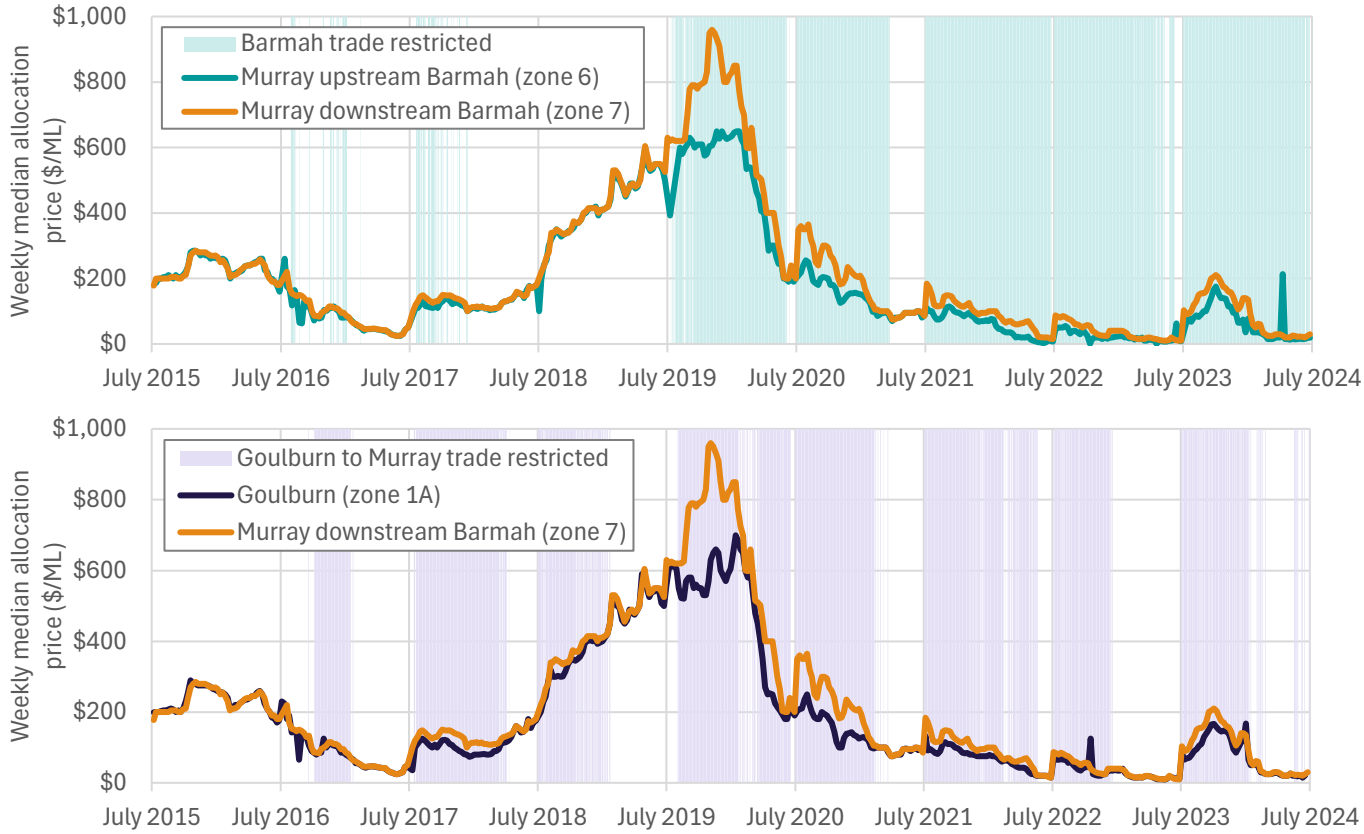
Two of the most influential restrictions on allocation trade in northern Victoria in recent years have been the limits on trade from upstream to downstream of Barmah and from the Goulburn to the Murray[10]. Data showing when these trade restrictions have been binding is presented in Figure 10 alongside median weekly allocation prices for the largest trading zones on each side of the restriction (Murray zone 6 and Goulburn zone 1A as the largest zones where trade is restricted from, and Murray zone 7 as the largest Victorian zone where is restricted to).

Figure 10 highlights a price divergence between the restricted zones whenever these limits were in effect, with high prices in the zone where trade is restricted into and lower prices in the zone where trade is restricted from. The divergence was most significant in 2019-20 when there was low water availability and high prices across northern Victoria and both limits remained closed for most of the year. In mid-November 2019 the median weekly price peaked at $960 per ML in the Murray zone 7 compared to $650 per ML in the Murray zone 6 and $660 per ML in the Goulburn zone 1A. These price differences reinforce the strong demand for water in the Murray downstream of Barmah and demonstrate the high willingness to pay from these water users.

[9] ***Goulburn to Murray trade review Long-term rules – fact sheet 1***

[10] In addition to the Barmah and Goulburn trade limits, other key trade restrictions that affect the northern Victorian market are limits on trade into Victoria from NSW, limits into NSW from Victoria and South Australia, and limits on trade out of the Murrumbidgee.

Figure 10: Allocation trade restrictions and price differentials



Notes:

1. *For the purpose of this graph, "trade restricted" means the opportunity to trade at the end of that day was 10 ML or less. This 10 ML value was adopted to avoid noise in the data resulting from small volumes of trade opportunity being available for periods of time when there were likely larger volumes market participants were seeking to trade.*
2. *Note that prior to November 2019 use from Goulburn tagged accounts was unrestricted.*
3. *Price data is median weekly data. Note that in some weeks there are very few trades, and prices may be skewed by a private lease arrangement or forward contact.*

Figure 10 also highlights that these trade restrictions have become more frequent over time. Trade downstream of Barmah restricted has been restricted most of the time since 2019-20, and trade out of the Goulburn has been restricted for large part of each year since 2017-18. The increasing restrictions reflects the strong demand for water in the Murray downstream of Barmah, including in NSW and South Australia as well as Victoria.

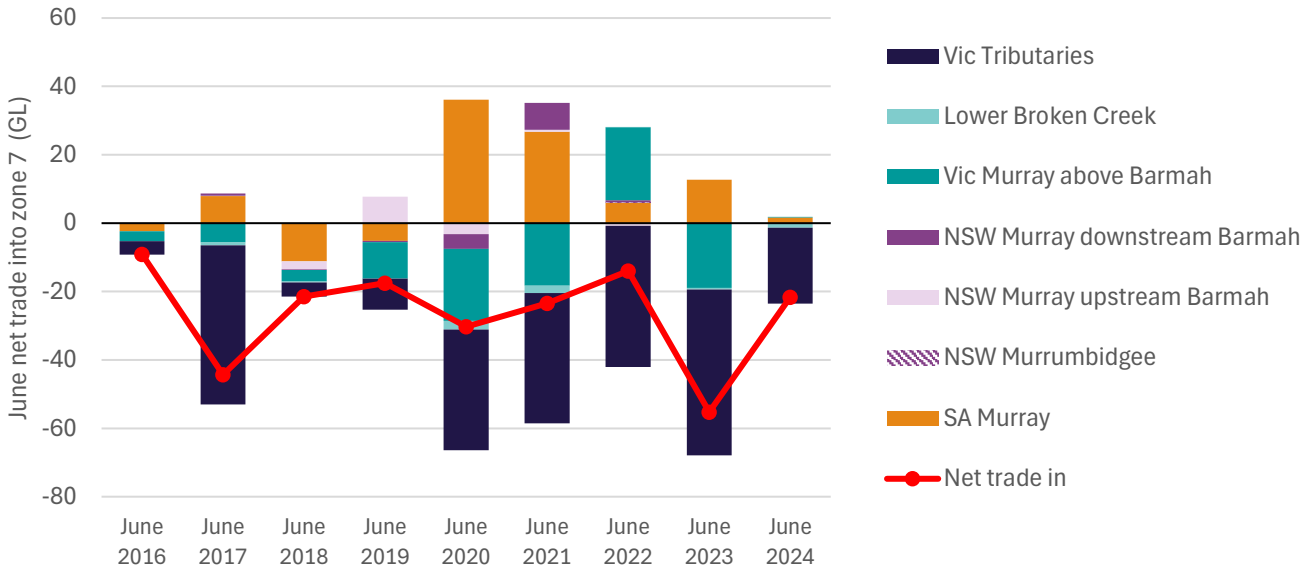
Most allocation trade under these limits now occurs as soon as trade opportunities become available. In practice, this means that most trade from upstream to downstream of Barmah occurs on 1 July, and most trade out of the Goulburn (since 2021 when the trade rule changed[11]) occurs on 1 July, 15 October and 15 December each year. Smaller volumes of allocation trade through the irrigation season occur when new opportunity is created by market participants trading in the opposite direction of the limit. These smaller volume trades are often taken up within minutes of the opportunity becoming available. As the irrigation season draws to a close, the trade limits have often opened up again due to net trade back upstream. This late season trend is explored further in the next section of this report.

[11] The timing of when Goulburn trade opportunities become available has shifted since July 2022 when changes were made to the Goulburn to Murray trade rule. Previously, the largest trade opportunities usually became available in summer and autumn when the Inter Valley Trade Account (which tracks the volume of water traded, but not yet delivered, from the Goulburn to the Murray system) fell below 200 GL. Under the new rule, most trade opportunity becomes available on 1 July, with additional trade opportunity released in October and December.

Late-season allocation trade

Even when the Barmah and Goulburn trade limits are closed for most of the year, they have often opened again late in the season due to net trade in the opposite direction of the limit. While this opening has occurred as early as April in some wet years, the restrictions have usually been opened in June. To provide an indication of the trend in late season allocation trade activity, Figure 11 presents net allocation trade into Murray zone 7 each June, including where that water is traded from (or to).

Figure 11: June net allocation trade into zone 7 (Murray downstream Barmah)



Notes:

1. *Victorian tributaries include trading zones 1A, 1B, 2, 3, 4A, 4C, and 5A. NSW upstream Barmah includes zones 10 and 10B. NSW downstream Barmah includes zones 11 and 14.*
2. *This chart excludes trade of environmental water. Environmental trade is defined as trades between environmental accounts, or $0 priced trades involving an environmental account.*

Figure 11 demonstrates that in June, the annual trend of net trade into Murray zone 7 (as presented in Figure 9), reverses with net trade out of the zone recorded each year. Net trade out of zone 7 in June has varied between 9 GL in 2016 and 55 GL in 2023. Water use is always very low in June, which suggests this late season trade trend is Murray zone 7 allocation holders seeking to reduce their risk of allocation write-off (forfeiture) before the end of the year by utilising available carryover space in other zones[12].

Since 2020 (when the Barmah and Goulburn trade limits started to be binding for most of the year), the largest proportion of the June trade out of Murray zone 7 has been into the Victorian tributaries, aligning where the largest volume of water shares are held. During this period there has been net trade in from interstate to zone 7 each June, highlighting that water share holders in the Murray zone 7 have also been realising the benefits of their carryover capacity by trading allocation in from across the southern connected Basin late in the season.

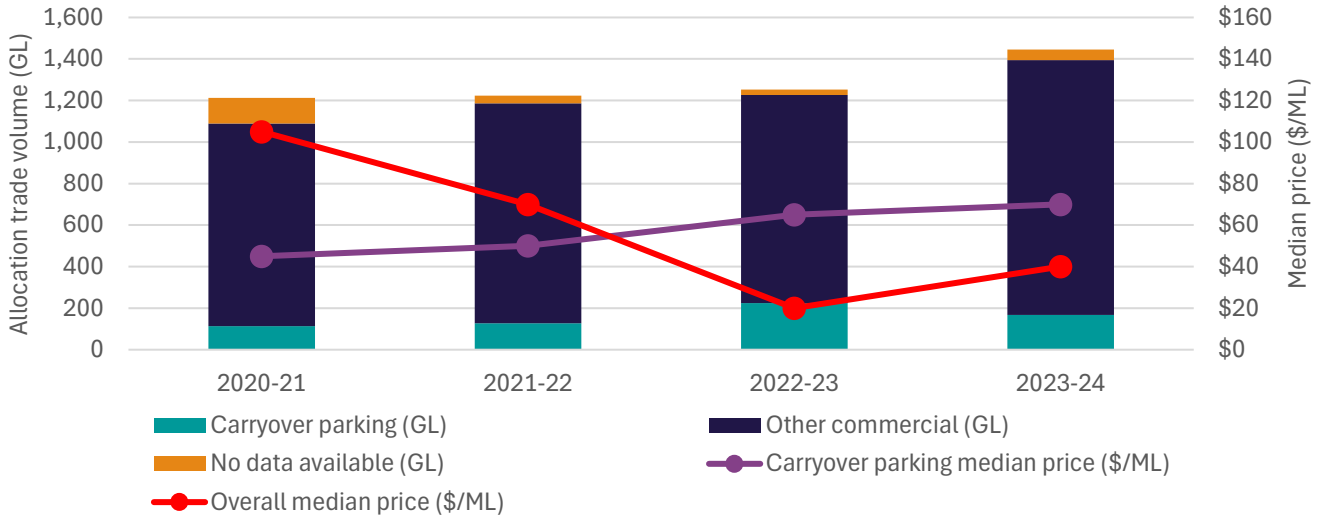
[12] Prior to the introduction of carryover in northern Victoria, any unused allocation remaining in accounts was written off on 30 June. This system had the unintended consequence of incentivising inefficient water use by driving irrigators to "use it or lose it" before the end of the current year. Now, subject to carryover rules, water share holders can carry over their unused water allocation into the following season. Victoria's carryover rules provide water share holders flexibility to manage their water security risks by building up their own drought reserves, improving certainty of early season access to water and minimising their write-off risks.

Carryover parking

Victorian carryover provisions have opened up opportunities for commercial arrangements where water share holders carry over water against their water share on behalf of someone else for a fee. This provides water share holders, water users and market participants more options and more flexibility to manage their water, and water share holders the opportunity to generate more income from their asset. This practice is commonly referred to as "carryover parking". Since August 2020, all online allocation trades have been reported with a 'reason for trade' including carryover parking (commercial trade to temporarily move water to another account to be held and carried over for the following year) and carryover parking return (the return trade in the new water year back into the original seller account).

Figure 12 draws on the four years of 'reason for trade' data to present how much commercial allocation trade has been for carryover parking compared to other reasons, and how the prices for carryover parking have differed from other commercial trade prices.

Figure 12: Commercial allocation trade for carryover parking compared to other reasons for trade



Notes:

1. *'Other commercial' allocation trade volumes refers to all commercial allocation trades recorded with a 'reason for trade' that is not carryover parking. This includes standard commercial trades, related party trades, private lease arrangements, allocation trades as part of a water share transfer, forward contracts and other. More information about these definitions is available at* ***Reason-for-trade-fact-sheet.pdf***
2. *'No data available' reflects commercial allocation trades recorded without a 'reason for trade'.*
3. *'Overall median price ($/ML)' in this chart excludes all trades with carryover parking as the 'reason for trade'.*

Carryover parking has comprised about 9-18% of the commercial allocation trades each year since 2020-21 (Figure 12). These trades mostly occur in June and include trades within and between trading zones. Increasing volumes of carryover parking between 2020-21 and 2022-23 suggest there has been increasing uptake as more people have become aware of this emerging market. The slight reduction in carryover parking volumes in 2023-24, when there was less 'unused' water available at the end of the year than in 2022-23, suggest that water availability at the end of the year may be a key driver influencing the volumes actually traded for the purpose of carryover parking.

The prices paid for carryover parking have gradually increased over the four years of data (from a median price of $45/ML in 2020-21 to a median price of $70/ML in 2023-24) indicating increased recognition of the value of carryover parking to water share holders and water users. The prices paid for carryover parking differ considerably from other commercial trades. In the two wetter years (2022-23 and 2023-24), when high water availability meant there was limited demand for within year water use, the median price for other commercial allocation trades was very low ($20/ML and $40/ML respectively), and carryover prices were higher ($65/ML and $70/ML).

6. Water market insights

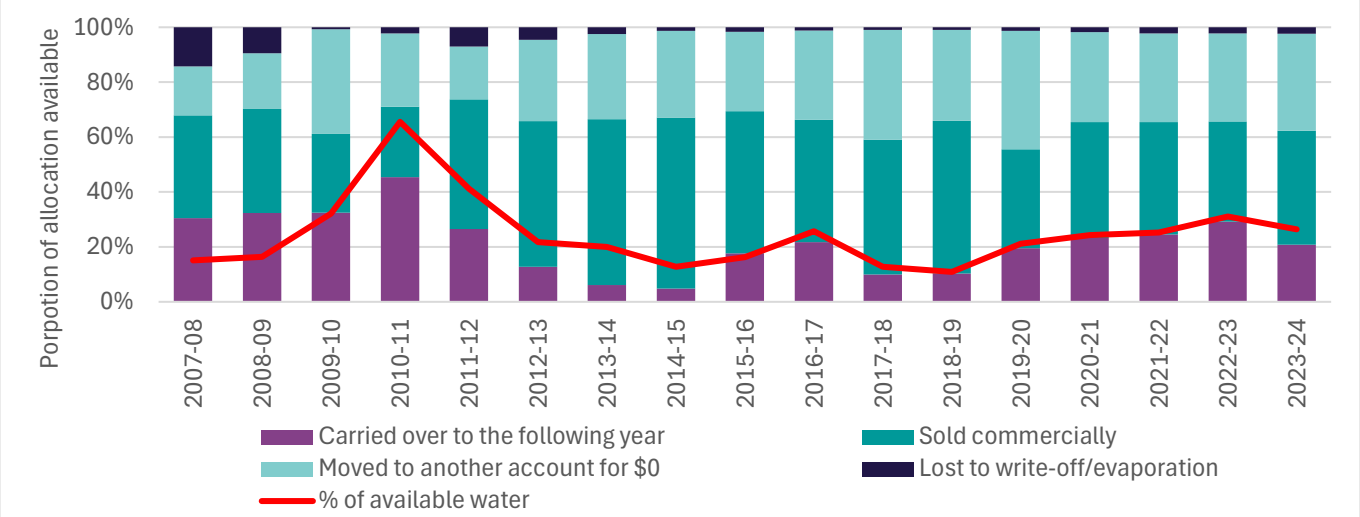
Water shares not linked to land

This section describes the trends in use of available water by holders of private water shares that are not linked to land. Allocation made to these water shares is either carried over or written off if it is not traded. Consistent with data presented in previous reports, holders of private water shares that are not linked to land tend to sell most of their available water commercially, making it available for irrigators to purchase and use (Figure 13). These holders (on average) also consistently move some allocation to another account for $0 in most years, which may reflect transfers into accounts for use on the account holders' own farm, or to a related party.

The proportion of available water that is carried over into the next year varies somewhat between years, but in recent years fairly closely mirrors the same statistic for private water users; that is, holders of water shares not linked to land tend to carry over about the same proportion of their available water that private water users do (Figure 13).

This data indicates that holders of water shares not linked to land are continuing to make most of the allocation they hold available for water users on the market.

Figure 13: How water available to holders of private water shares not linked to land was used



Notes:

1. *The 'allocations available' in this data is the sum of all available water per year. This includes new season allocations, trade in from other accounts, and allocation carried over from the previous year adjusted for spills.*
2. *'Sold commercially' reflects the total volume of commercial trades out of privately held accounts with water shares not linked to land.*
3. *"Moved to another account for $0' reflects the total volume of $0 priced allocation trades out of privately held accounts not linked to land.*
4. *This data differs from equivalent reporting in the 2018 Market Trends Report which was based on water shares not tied to land, rather than not linked to land. Further analysis in how these reporting methods differ is presented in Appendix 1.*

Water users' approach to securing water allocations

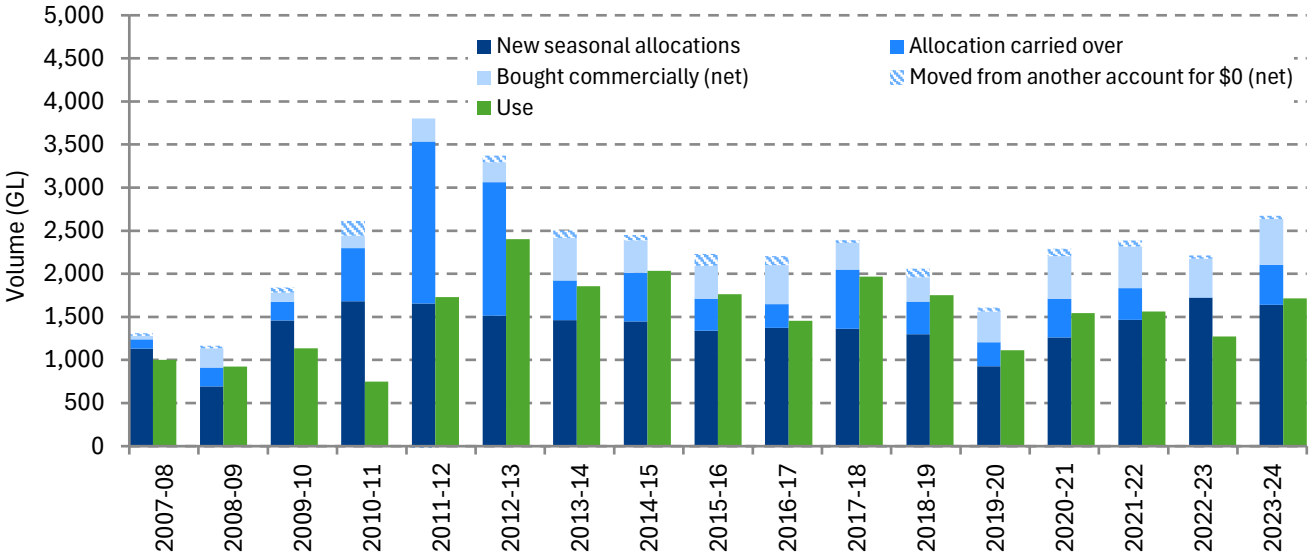
Private water users – as distinct from holders of water shares not linked to land mostly rely on seasonal allocations directly from water shares as their main mechanism for accessing water; however in most years this is not sufficient to cover all their water use needs and they also rely on carryover and/or trade (Figure 14).

In average and wet years, the allocation carried over by private water users along with their seasonal allocations has been sufficient to cover their annual water needs. Large carryover volumes into 2011-12 and 2012-13 could have supported exceptionally high water use, and while there was strong water use in those years, there were also large

carryover reserves at the end of those years that were run down over the subsequent three years to more typical levels. Carryover rules have since changed[13], as has user behaviour. In drier years, such as 2018-19, carryover has helped bolster private water use to above average levels. In more recent wet years, carryover available for use has been impacted by spills from storages. In 2022-23 all allocation carried over from the previous year in Victorian Murray, Goulburn and Campaspe accounts spilled, meaning these customers had no access to additional carryover beyond their water share volume in these systems for that year. However, the unavailability of carryover is unlikely to have constrained water use in 2022-23, given full allocations in 2022-23 and the impacts of flooding, on-farm rainfall and relatively mild summer conditions suppressing irrigation demand.

As a group, private water users in northern Victoria have consistently been net buyers of allocation – trading in water from other states, water corporations, environmental water holders and other private holders who haven't used water. The reliance on trade by this group becomes more important in dry years, such as 2018-19 when allocations and carryover were not sufficient to meet the group's water needs. In 2023-24, they traded in 566 GL or 33% of their water use requirements, most of which (528 GL) was commercial trade. Every year this group of water users moves some allocation into their accounts via $0 priced trades. These $0 priced trades reflect users moving water between their accounts (or accounts of related parties) and can include moving water from accounts with water shares not linked to land or interstate entitlements to accounts linked to land.

Figure 14: Private water users – where they source their water from



Notes:

1. *Only privately held northern Victorian accounts with water use recorded in the relevant year are included in this data.*
2. *'Allocation carried over' reflects the carried over water available after deductions from spillable water accounts.*
3. *'Bought commercially' reflects the net volume of commercial trades into private water user accounts.*
4. *'Moved from another account for $0' reflects the net volume of $0 priced trades into private water user accounts.*
5. *There was no net trade out in any year.*

LMW diverters are the private water user group most dependent on trade to support their water use. In 2023-24, new seasonal allocations (237 GL) and allocation carried over from the previous year (20 GL) accounted for only 61% of LMW diverters' use (compared to 140%, 223% and 103% of use for GMW districts, GMW diverters and LMW districts respectively). The latter three groups have enough allocations to cover their needs in most years, with carryover

[13] Changes from the 2012 Carryover Review, including most notably a limit on carryover to the volume of the entitlement in Murray, Goulburn and Campaspe systems were implemented in 2013 and 2014. More information about this review and subsequent changes is available at ***https://waterregister.vic.gov.au/water-entitlements/carryover/carryover-review-2012***.

providing additional support in low allocation years, while LMW diverters as a group have not met their combined use requirements without trade since the 2012-13 year. These differing regional dependencies likely reflect that more LMW diverters are relatively new entrants who have developed business models based around a range of different products for accessing water, including entitlement and allocation trade through different means such as forward contracts and private leases. This is supported by the first four years of 'reason for trade' data: over this period, LMW diverters reported almost 9% of their allocation trades as either a forward contract or private lease, compared to 3%, 1% and 1% for GMW district, GMW diverter and LMW district users respectively.

Frequent Traders

To increase market transparency, Victoria has commenced reporting on 'frequent traders' in the Goulburn and Murray systems. A frequent trader is defined as the owner of a Victorian allocation account that has undertaken more than 20 commercial allocation trades, within, to, or from the Goulburn and Murray systems during the year excluding interstate trades[14]. Quarterly updates identifying these holders are now published on the Victorian Water Register website, ***linked here***.

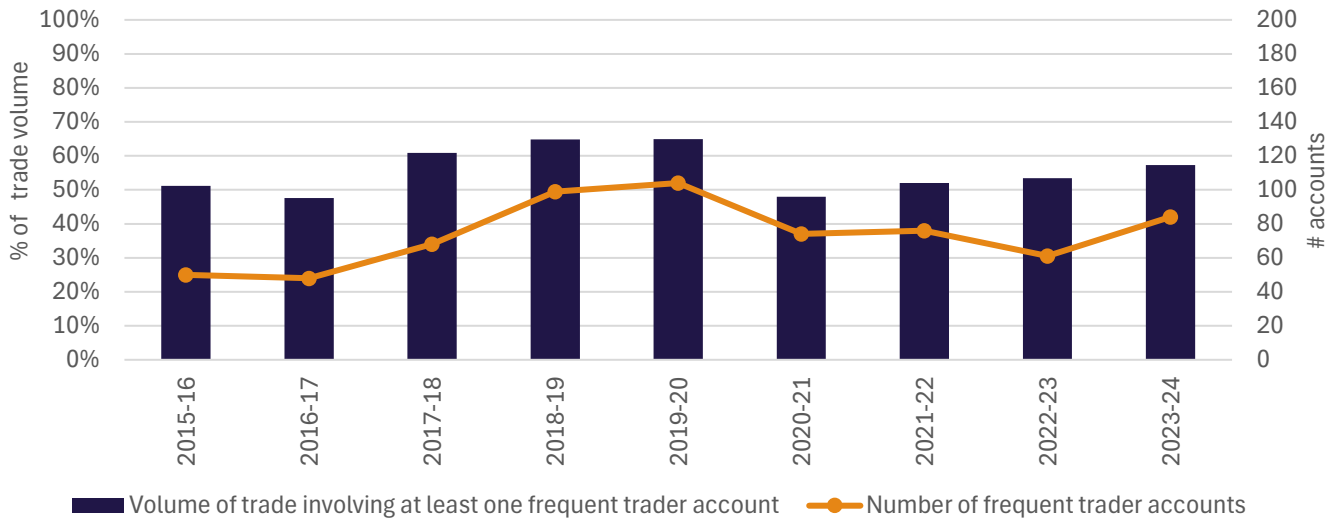
Qualifying to be on the frequent trader list does not indicate any wrongdoing. Many parties rely on water allocation markets as part of their business model. The list simply provides transparency about account holders who trade relatively frequently in the Goulburn and Murray systems, as these traders have the potential to have more influence in the market.

Analysis shows that since 2015-16 there have been between 48 and 104 frequent trader accounts each year (Figure 15). This number was higher in dry years when prices were higher (2018-19 and 2019-20) and lower in the wet, low-price years (such as 2022-23). These fluctuations are broadly in line with fluctuations in the total number of commercial trades, which was also higher in those drier years. It is worth noting that when allocation prices were highest in 2019-20, frequent traders sold a smaller proportion (33%) of the total commercial trade than in years with lower prices (including 2022-23 (40%) and 2023-24 (42%).

Between 25% and 42% of the total volume of commercial trade in the Victorian Murray and Goulburn has been sold from frequent trader accounts each year since 2015-16. This is notable given that frequent trader accounts make up approximately 1% of the total number of accounts trading each year (Table 1). When counting any trades where a frequent trader is the buyer *or* the seller (i.e. trades involving at least one frequent trader account), this rises to 48 to 65% of the total volume (Figure 15). There is no clear trend across the time period. The high proportion can be explained by two factors: 1. by definition, these are accounts that trade more frequently than all other accounts; 2. the average parcel size of trades by frequent trader accounts is approximately twice as large (164 ML) as the average parcel size of all other trades (85 ML).

[14] Victoria does not have information on what happens to the water when it is traded interstate, as the interstate party is not recorded.

Figure 15: Proportion of Goulburn and Murray commercial allocation trade involving at least one frequent trader account



Notes:

1. *The total trade volume here refers to all trades that meet the criteria used to define frequent traders. That is: approved, commercial trades, within, to, or from the Goulburn and Murray systems (zones 1A, 1B, 3, 6, 6B, and 7).*

Table 1: Number of unique allocation accounts involved in commercial trade in the Goulburn and Murray

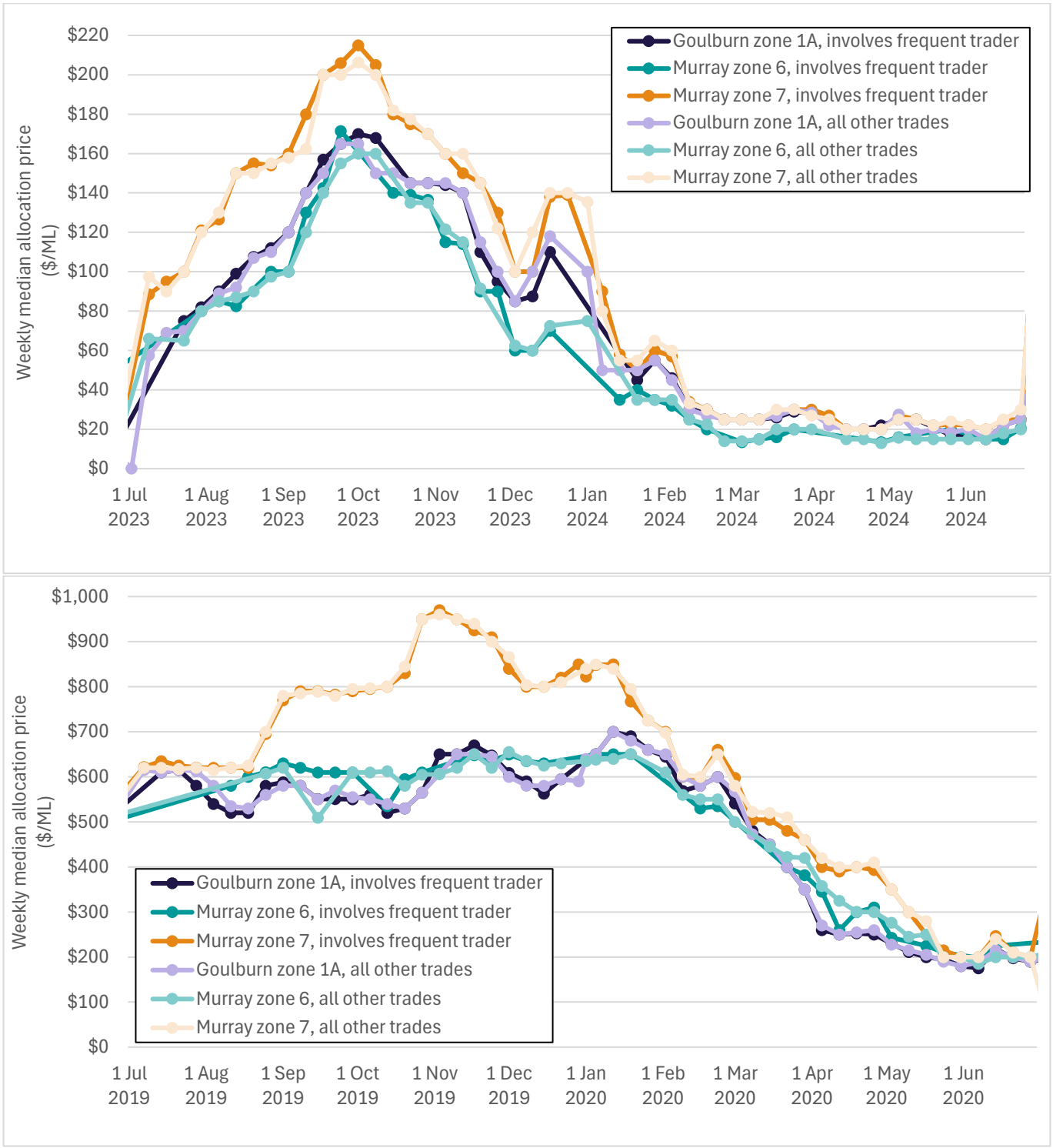
Water year	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24
Frequent trader accounts	50	48	68	99	104	74	76	61	84
All other trader accounts	6,844	5,540	7,425	8,617	7,249	6,919	6,364	5,188	6,465
Total unique accounts trading	**6,894**	**5,588**	**7,493**	**8,716**	**7,353**	**6,993**	**6,440**	**5,249**	**6,549**

Notes:

1. *The other trader accounts identified here are all accounts that have made at least one trade that meets the criteria used to define frequent traders. That is: approved, commercial trades, within, to, or from the Goulburn and Murray systems (zones 1A, 1B, 3, 6, 6B, and 7).*

A comparison of the weekly median price for trades involving a frequent trader with all other trades (not involving a frequent trader) in northern Victoria since 2015-16 show no systematic differences in prices between the two groups. Figure 16 shows that when controlling for trading zone, the weekly median allocation prices for the two groups are closely aligned in both the recent year of 2023-24 and in 2019-20 as an example of a dry year when allocation prices were high. Prices were similarly aligned for all other years between 2015-16 and 2023-24.

Figure 16: Weekly median allocation price for trades involving at least one frequent trader, compared to all other trades – 2023-24 and 2019-20



Notes:

1. *Price data is median weekly data. The data is calculated based on all approved, priced trades with a source trading zone of the category indicated.*
2. *The median price data includes only weeks with at least 5 trades for the given category.*
3. *Data for 2023-24 excludes all trades with a 'reason for trade' of 'Private lease arrangement (not LTT)' or 'Forward contract'. The 2020-19 year was prior to the introduction of 'reason for trade' requirements and therefore does not exclude these types of trades.*

7. Conclusions

This report provides an update since the last Water Market Trends report (DEECA 2019), with a primary focus on new data and trends since 2017-18, in the context of a longer history of Victoria's northern water markets. The report provides new data and discussion, both on topics from the previous report but also emerging issues and topics of interest to market participants.

Private rural water use in GMW districts has continued to account for the largest segment of private use in northern Victorian declared river systems, but also the most variable year to year. The trend identified in the last version of this report of increasing total use by the group of LMW diverters has stabilised; looking at the longer period of record there is no material trend evident since 2012-13.

A look at water share trade median prices has shown a significant diversion of Murray downstream of Barmah (Zone 7) water share prices above other trading zones since 2018, likely reflecting the growth of perennial horticulture demand in that region and increasing allocation trade restrictions into the region.

Water allocation trade overall volumes continued to increase year on year. The LMW diverter group is the private water user group most dependent on trade to meet their water needs.

Updated information is also provided on the behaviour of holders of water shares not linked to land; i.e. what they do with their allocation each year.

Finally, two key improvements in market transparency, the collection of 'reason for trade' data and a new analysis on market participants who meet the criteria for 'frequent traders', have both provided new insights to this report. 'Reason for trade' helps to characterise trends such as the emergence of a market for carryover parking. The availability of this data also improves other analyses such as allocation price trends. Data showing the behaviour of frequent traders in the market highlights that while this group of organisations and individuals is involved in an outsized proportion of traded volume, there is no evidence of systematic differences in the price of these trades compared to other, less frequent, traders.

8. References

Bureau of Meteorology (2023), Australian Water Markets Report 2021-22.

Department of Energy, Environment and Climate Action (2023), Victorian Water Trading: 2022-23 Annual Report ***Victorian Water Trading***

Department of Environment, Land, Water and Planning (2016), Water Market Trends: Trends in Northern Victorian Water Trade 2001-2015 ***Microsoft Word - Water market trends 20160224.docx***

Department of Environment, Land, Water and Planning (2019), Water Market Trends: Updated Trends in Northern Victorian Water Trade 2001-2018 ***Water-Market-Trends-Update-2018_web.pdf***

Department of Natural Resources & Environment (2001), The Value of Water: a guide to water trading in Victoria (accessible at ***Trade reports - Water Register***)

Goulburn Broken Catchment Management Authority (2023). Regional Irrigated Land and Water Use Mapping in the Goulburn Murray Irrigation District - Fact Sheets - 2021/2022. ***Goulburn Murray Irrigation District 2021-22 snapshot***

Goulburn Broken Catchment Management Authority (2024), Land and Water Use Mapping in Regulated Diversions Catchments of Northern Victoria Technical Report 2022/23 ***Land and Water Use Mapping report***

Mallee Catchment Management Authority (2022). 2021 Mallee Horticulture Crop Report. ***2021 Mallee Horticulture Crop Report***

SunRISE (2022), Irrigated crop area data for the Lower Murray-Darling 2003 to 2021 ***Irrigated crop area data for the lower Murray Darling 2003 to 2021***

SunRISE (2023), Milestone 5 Report: Supply mapping of expansion areas since 2021 ***Project proposal: Irrigated crop area data 2021 to 2027***

Appendix 1 – Water shares linked to land

A new method for reporting on water shares "linked to land" has been introduced in this report in response to changes to the *Water Act 1989*. The new method better reflects where allocations made to water shares are approved for use on land. This Appendix explains the legislative changes, the new reporting methods, and how they compare to previous methods.

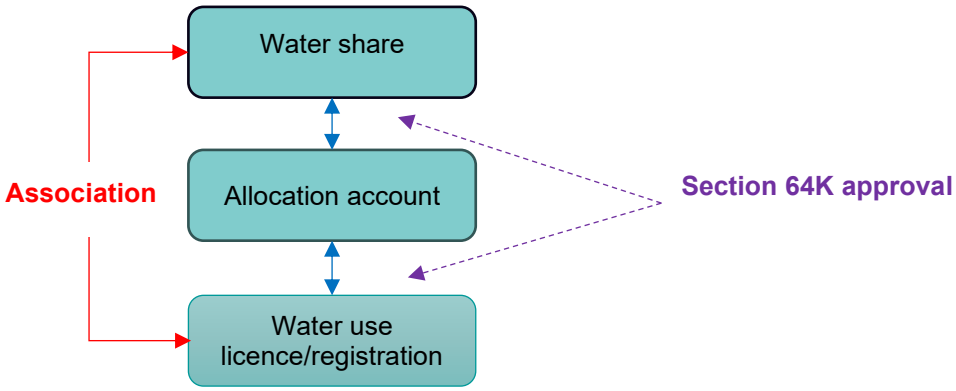
Legislative changes – removal of association

Changes to the *Water Act 1989* were introduced on 20 November 2023 which removed the ability to associate a water share with land (under Part 3A, Division 10). Association was one of the ways that the water under a water share could be approved to be used on land, and was originally put in place to manage the following limits:

- no more than 10% of the water share volume in a system could be owned without being linked to land (this limit was abolished in 2009), and
- no more than 4% of the water share volume in an irrigation area could be moved out the area in a year (this limit was abolished in 2014).

When unbundling occurred in northern Victoria (1 July 2007), all water shares were associated with land. Holders could choose to disassociate their water shares from land (and associate it with other land if they chose to) subject to the above limits (when they were in effect).

Association was reflected in the Victorian Water Register by a direct connection between the water share and the water-use licence or registration on the relevant land (refer red line in figure below). Associating a water share provided approval to use water allocated to the water share on the specified land. However, such approval could also be obtained via an approval under section 64K of the *Water Act 1989.* Section 64K approvals are represented in the Victorian Water Register by a connection between a water share and an allocation account with a connected water-use licence or registration (refer blue lines in figure below).



Over time, water users increasingly opted to rely on section 64K approvals for their approvals to use water. There were different fees for water shares associated with land, so over time more and more people chose to link their water share to land without formally associating it under Part 3A, Division 10 of the *Water Act 1989*. As such, reporting on water shares associated with land increasingly became a poor measure of where water shares were approved to be used.

The legislation changes made in 2023 simplified the approvals required to use allocation at a particular place. This included removing the unnecessary ability to associate water shares with land – reflecting the fact that association was no longer required to manage the 10% and 4% limits and was no longer a meaningful indication of where water share allocations were planned to be used. The new approach to approvals to use water share allocations are now consistent – with all water users (allocation account holders with a linked water-use licence/registration) having a *general place of take approval* issued under Part 4AA for every place (service point) they take water.

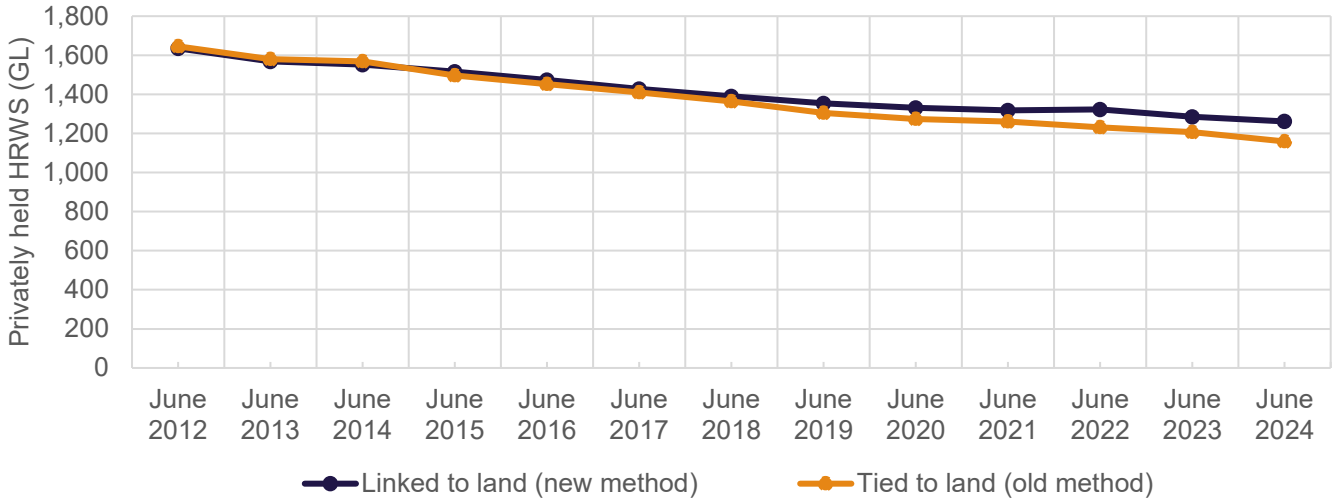
New reporting method – water shares linked to land

Previous market trend reporting (in 2016 and 2019) reported on water shares "tied to land" based on whether the water shares were "associated" to a water-use licence/registration. This report uses a new method to report on water shares "linked to land" based on whether a water share is linked to an allocation account with a general place of take approval (i.e. the allocation account has a link to a water-use licence/registration).

Similar to previous reports, the data reported for each water year is based on the active links recorded in the Victorian Water Register on 30 June of each year.

Figure 17 compares these two reporting methods. It shows the volume of high-reliability water shares 'linked to land' (new method) and the volume 'tied to land' (old method). The old method of reporting was not an accurate picture of the volume of water shares linked to land as more water share holders chose to disassociate their water shares from land, even though they retained other approvals to use that water on land. The inaccuracy became more significant over time. The new method provides a more reliable indication of when water shares are linked to land, and aligns with the 2023 legislation changes, and the single approach to approvals to use water on land going forwards.

Figure 17: Comparison of calculation methods - privately held high-reliability water share linked/tied to land on 30 June



New reporting method – location of take under water shares

Previous market trend reporting (in 2016 and 2019) reported on the location of water shares "tied to land" based on the delivery system recorded on the water share. This is no longer appropriate because the delivery system on the water share was derived from an associated water-use licence/registration, and as discussed above, this data was becoming increasingly inaccurate. Additionally, now that the ability to associate a water share has been removed, the delivery system data on a water share is no longer being maintained.

A new approach to reporting on where a water share is approved to be used is adopted for this report, that derives the delivery system from the water-use licence/registrations linked to the water share's allocation account. Based on 2023-24 data, most high-reliability water shares are linked to a single delivery system. However, where a water share's allocations are approved for take at multiple places we have made some simplifying assumptions to enable reporting on the location of take. Where a water share is linked to multiple delivery systems, the water share is reported as being located in the delivery system where the most water has been used in the last two years.[15]

[15] If in the preceding two years, the water share's allocation account had equally high proportions of water use in two delivery systems, then the lowest numbered delivery system is reported as the water share location.

Appendix 2 – Detailed data on water trade and use

Figure 18: Map of irrigation districts in northern Victoria

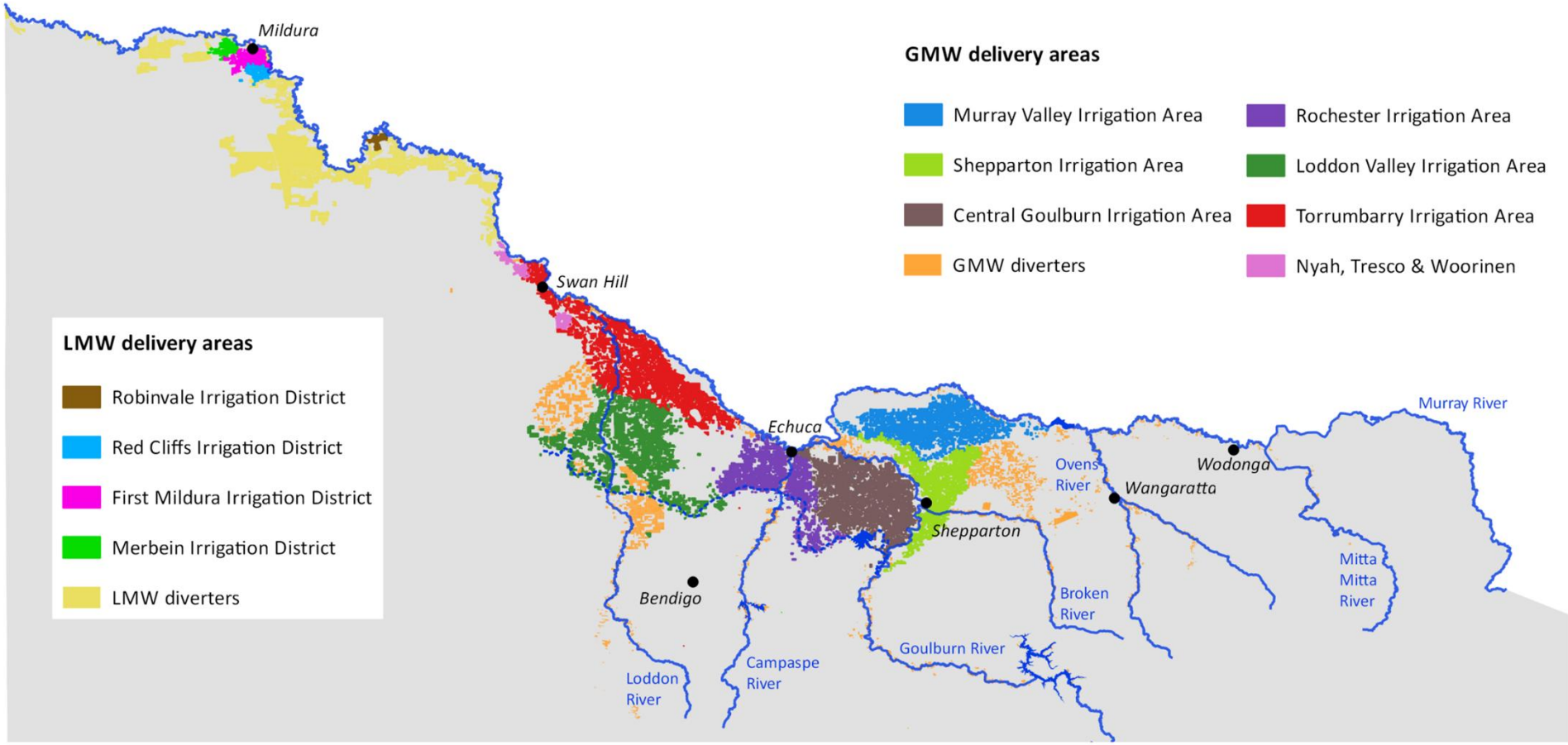


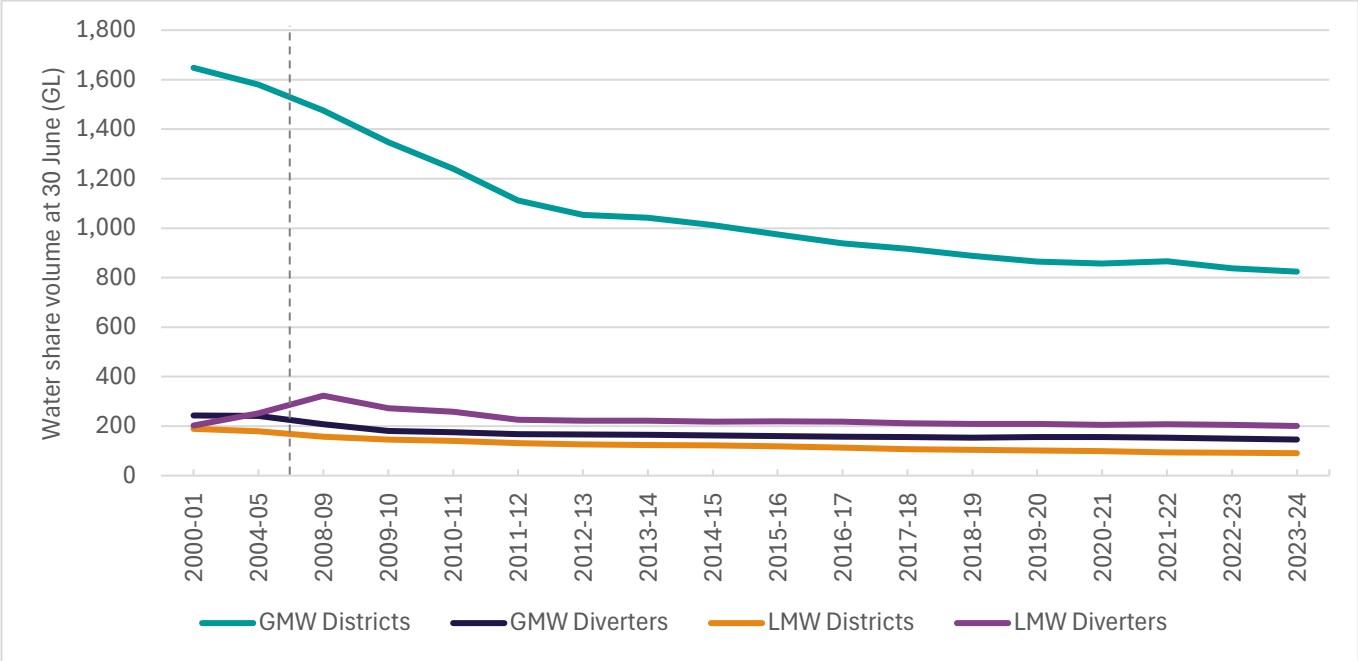
Table 2: Northern Victorian high-reliability water share volume at 30 June

Water year	Environmental water holder		Water corporation		Private - not linked to land		Private - linked to land		Total
	GL	%	GL	%	GL	%	GL	%	
2000-01	0	-	0	-	0	-	2,283	100%	2,283
2004-05	0	-	0	-	0	-	2,251	100%	2,251
2008-09	0	1%	0	2%	125	12%	2,163	85%	2,288
2009-10	160	7%	46	2%	128	6%	1,947	85%	2,280
2010-11	283	12%	64	3%	120	5%	1,813	80%	2,279
2011-12	463	20%	67	3%	117	5%	1,635	72%	2,281
2012-13	499	22%	74	3%	142	6%	1,569	69%	2,285
2013-14	539	23%	59	3%	146	6%	1,552	68%	2,295
2014-15	605	26%	62	3%	156	7%	1,516	65%	2,338
2015-16	648	27%	62	3%	197	8%	1,473	62%	2,380
2016-17	652	27%	63	3%	237	10%	1,428	60%	2,380
2017-18	661	28%	63	3%	272	11%	1,390	58%	2,386
2018-19	732	30%	61	2%	307	13%	1,355	55%	2,455
2019-20	732	30%	62	3%	330	13%	1,331	54%	2,455
2020-21	733	30%	62	3%	344	14%	1,317	54%	2,456
2021-22	733	29%	48	2%	397	16%	1,322	53%	2,500
2022-23	739	29%	48	2%	435	17%	1,285	51%	2,507
2023-24	747	30%	50	2%	456	18%	1,262	50%	2,515
Change from 2018 to 2024	**86**	**2%**	**-12**	**-1%**	**184**	**7%**	**-129**	**-8%**	**129**

Notes:

1. *Refer to Appendix 1 for details about how "linked to land" data differs from "tied to land" data reported in the 2018 report.*
2. *Volumes at 30 June 2001 and 2005 reflect the volume of water rights held (since water shares had not yet been introduced). 30 June 2009 values are as reported in the 2018 trends update due to difficulties of working with the historical data.*

Figure 19: Changes in private high-reliability water shares linked to land in northern Victoria



Notes:

1. *Values prior to 2009-10 are volumes as reported from the 2018 trends update.*
2. *The vertical dotted line indicates that 2000-01 and 2004-05 are not continuous with the rest of the years included.*

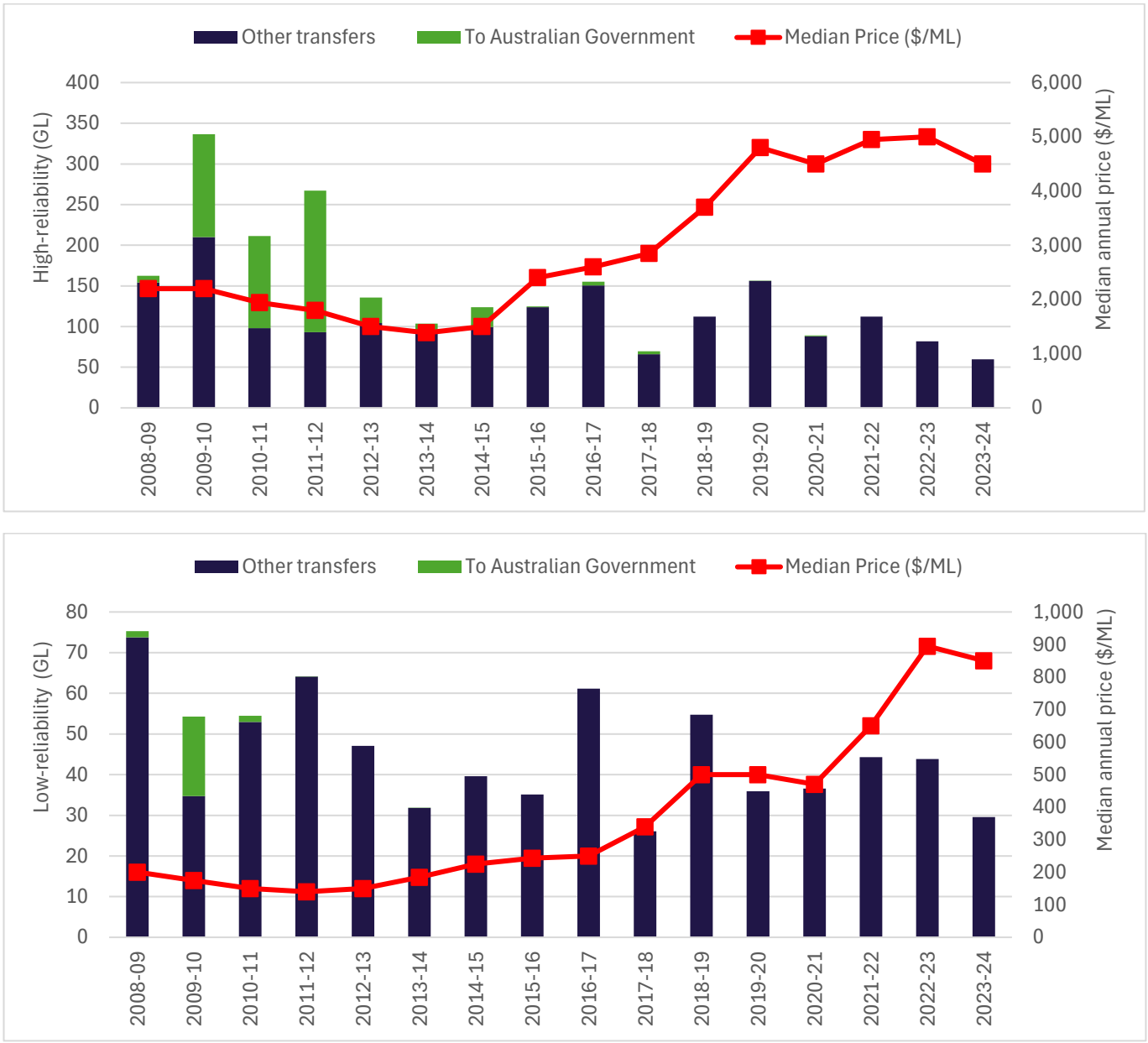
Table 3: Changes in private high-reliability water shares linked to land in northern Victoria

Water year	GMW Districts (GL)	GMW Diverters (GL)	LMW Districts (GL)	LMW Diverters (GL)	Total
2000-01	1,648	243	189	203	2,283
2004-05	1,580	241	179	252	2,252
2008-09	1,475	208	157	323	2,163
2009-10	1,348	180	146	273	1,947
2010-11	1,240	175	140	258	1,813
2011-12	1,111	167	131	226	1,635
2012-13	1,054	167	126	222	1,569
2013-14	1,041	165	124	221	1,552
2014-15	1,012	163	123	218	1,516
2015-16	975	160	119	219	1,473
2016-17	939	158	114	218	1,428
2017-18	916	156	107	211	1,390
2018-19	888	154	104	209	1,355
2019-20	864	156	102	209	1,331
2020-21	857	156	99	205	1,317
2021-22	866	153	94	208	1,322
2022-23	837	149	93	205	1,285
2023-24	824	146	91	201	1,262

Notes:

1. *Values prior to 2009-10 are volumes as reported from the 2018 trends update.*

Figure 20: Transfers and price of high and low-reliability water shares in northern Victoria



Notes:

1. *The volume of water shares transferred includes transfers of ownership recorded during each year. Transfers of ownership may or may not involve a change in location of water shares.*
2. *The median price is an annual median for all northern Victorian water shares based on all transfers of ownership recorded in the year with non-zero prices.*

Table 4: Transfers and price of high-reliability water shares in northern Victoria

Water year	Volume of high-reliability water shares transferred (GL)		Median Price ($/ML)
	To Australian Government	Other transfers	
2008-09	8	154	2,175
2009-10	127	210	2,200
2010-11	113	98	1,940
2011-12	174	93	1,800
2012-13	31	104	1,500
2013-14	10	93	1,380
2014-15	25	99	1,500
2015-16	1	124	2,400
2016-17	5	151	2,600
2017-18	3	66	2,850
2018-19	-	112	3,700
2019-20	0	156	4,800
2020-21	1	88	4,500
2021-22	-	112	4,950
2022-23	-	82	5,000
2023-24	-	60	4,500

Notes:

1. *The volume of water shares transferred includes transfers of ownership recorded during each year. Transfers of ownership may or may not involve a change in location of water shares.*
2. *The annual median price is based on all northern Victorian transfers recorded during the year with non-zero prices.*

Table 5: Transfers and price of low-reliability water shares in northern Victoria

Water year	Volume of low-reliability water shares transferred (GL)		Median Price ($/ML)
	To Australian Government	Other transfers	
2008-09	1	74	200
2009-10	20	35	175
2010-11	1	53	150
2011-12	0	64	140
2012-13	-	47	150
2013-14	0	32	185
2014-15	-	40	225
2015-16	-	35	244
2016-17	-	61	250
2017-18	-	26	340
2018-19	-	55	500
2019-20	-	36	500
2020-21	-	37	470
2021-22	-	44	650
2022-23	-	44	895
2023-24	-	30	850

Notes:

1. *The volume of water shares transferred includes transfers of ownership recorded during each year. Transfers of ownership may or may not involve a change in location of water shares.*
2. *The annual median price is based on all northern Victorian transfers recorded during the year with non-zero prices.*

Table 6: Volumes of high-reliability water share tied to land compared to annual water use

Water year	GMW districts			GMW Diverters			LMW districts			LMW Diverters			Total		
	HRWS (GL)	Use (GL)	%	HRWS (GL)	Use (GL)	%	HRWS (GL)	Use (GL)	%	HRWS (GL)	Use (GL)	%	HRWS (GL)	Use (GL)	%
2004-05	1,580	1,686	107%	241	133	55%	179	136	76%	252	196	78%	2,252	2,151	96%
2008-09	1,475	556	38%	208	43	21%	157	90	57%	323	237	73%	2,163	925	43%
2009-10	1,348	745	55%	180	33	18%	146	87	59%	273	270	99%	1,947	1,135	58%
2010-11	1,240	495	40%	175	19	11%	140	46	33%	258	190	74%	1,813	751	41%
2011-12	1,111	1,268	114%	167	59	35%	131	94	72%	226	310	137%	1,635	1,732	106%
2012-13	1,054	1,777	169%	167	104	62%	126	113	89%	222	408	184%	1,569	2,402	153%
2013-14	1,041	1,316	126%	165	89	54%	124	98	79%	221	354	160%	1,552	1,857	120%
2014-15	1,012	1,450	143%	163	90	55%	123	96	78%	218	397	182%	1,516	2,033	134%
2015-16	975	1,154	118%	160	82	52%	119	104	88%	219	422	192%	1,473	1,762	120%
2016-17	939	928	99%	158	66	42%	114	89	79%	218	371	170%	1,428	1,454	102%
2017-18	916	1,338	146%	156	96	62%	107	104	97%	211	429	203%	1,390	1,967	142%
2018-19	888	1,090	123%	154	89	58%	104	112	108%	209	460	220%	1,355	1,752	129%
2019-20	864	531	61%	156	52	33%	102	98	96%	209	432	207%	1,331	1,113	84%
2020-21	857	940	110%	156	71	46%	99	98	99%	205	436	212%	1,317	1,545	117%
2021-22	866	953	110%	153	68	45%	94	95	101%	208	448	215%	1,322	1,564	118%
2022-23	837	814	97%	149	55	37%	93	77	82%	205	327	160%	1,285	1,273	99%
2023-24	824	1,127	137%	146	77	52%	91	95	104%	201	418	208%	1,262	1,716	136%

Notes:

1. *High-reliability water share volumes prior to 2009-10 are volumes as reported from the 2018 trends update. Water use volumes for 2004-05 are as reported from the 2018 trends update.*

Table 7: Volumes of high-reliability water share tied to land compared to annual water use - GMW districts

Water year	GMW Districts																						
	Central Goulburn Irrigation Area			Loddon Valley			Murray Valley Irrigation Area			Nyah, Tresco, Woorinen			Rochester Irrigation Area			Shepparton Irrigation Area			Torrumbarry Irrigation Area			Campaspe Irrigation Area		
	HRWS (GL)	Use (GL)	%	HRWS (GL)	Use (GL)	%	HRWS (GL)	Use (GL)	%	HRWS (GL)	Use (GL)	%	HRWS (GL)	Use (GL)	%	HRWS (GL)	Use (GL)	%	HRWS (GL)	Use (GL)	%	HRWS (GL)	Use (GL)	%
2004-05	373	389	104%	216	222	103%	257	283	110%	31	20	65%	182	199	109%	174	157	90%	327	405	124%	19	10	53%
2008-09	342	140	41%	196	70	36%	250	92	37%	29	17	58%	167	58	35%	161	64	40%	312	112	36%	18	3	18%
2009-10	316	171	54%	166	100	60%	229	127	55%	28	17	61%	153	84	55%	152	72	48%	290	173	60%	14	0	2%
2010-11	297	121	41%	153	44	29%	215	92	43%	30	10	35%	141	52	37%	143	42	29%	260	130	50%	1	3	633%
2011-12	259	289	112%	136	152	112%	195	210	108%	29	18	62%	124	153	123%	132	103	79%	236	332	140%	0	11	-
2012-13	244	414	170%	128	234	183%	182	302	166%	29	23	79%	119	215	180%	126	163	130%	225	423	188%	0	1	-
2013-14	238	313	131%	124	164	132%	174	222	127%	26	21	79%	117	163	139%	122	127	104%	239	304	127%	0	2	-
2014-15	234	347	148%	123	204	166%	167	227	136%	42	23	54%	113	173	153%	118	128	108%	214	348	162%	0	1	-
2015-16	226	277	123%	121	155	128%	159	194	122%	36	22	60%	110	150	137%	113	108	95%	210	246	117%	0	1	-
2016-17	219	212	97%	118	112	95%	149	168	112%	23	19	83%	105	109	104%	110	85	77%	213	221	104%	0	1	-
2017-18	212	310	146%	116	185	159%	143	231	161%	35	23	65%	103	167	162%	106	128	121%	200	293	146%	0	0	-
2018-19	206	266	129%	116	142	123%	138	170	123%	27	22	83%	101	141	140%	103	117	113%	196	232	118%	0	0	-
2019-20	203	138	68%	114	71	62%	132	74	56%	26	19	73%	99	63	63%	100	63	63%	191	103	54%	0	0	-
2020-21	198	214	108%	114	151	133%	131	128	98%	22	21	97%	99	128	130%	99	88	90%	196	209	107%	0	0	-
2021-22	207	232	112%	121	144	120%	131	127	97%	20	19	93%	102	138	135%	102	77	76%	184	216	118%	0	0	-
2022-23	206	189	92%	119	111	93%	129	138	107%	21	14	68%	99	104	105%	98	73	74%	166	186	112%	0	0	-
2023-24	201	252	126%	116	159	138%	128	205	160%	20	17	81%	101	144	142%	96	98	102%	162	251	155%	0	0	-

Notes:

1. *High-reliability water share volumes prior to 2009-10 are volumes as reported from the 2018 trends update. Water use volumes for 2004-05 are as reported from the 2018 trends update.*

Table 8: Volumes of high-reliability water share tied to land compared to annual water use - LMW districts

Water year	LMW Districts								
	First Mildura Irrigation District			Robinvale, Red Cliffs, Merbein			LMW districts subtotal		
	HRWS (GL)	Use (GL)	%	HRWS (GL)	Use (GL)	%	HRWS (GL)	Use (GL)	%
2004-05	73	50	68%	106	86	81%	179	136	76%
2008-09	63	30	48%	94	60	64%	157	90	57%
2009-10	61	30	50%	85	56	66%	146	87	59%
2010-11	59	13	22%	81	33	41%	140	46	33%
2011-12	56	28	50%	75	66	88%	131	94	72%
2012-13	53	39	74%	73	73	100%	126	113	89%
2013-14	52	34	65%	72	64	89%	124	98	79%
2014-15	52	33	63%	71	63	89%	123	96	78%
2015-16	51	36	69%	68	68	101%	119	104	88%
2016-17	49	30	61%	64	59	92%	114	89	79%
2017-18	47	36	75%	60	69	115%	107	104	97%
2018-19	46	39	85%	58	73	125%	104	112	108%
2019-20	44	34	78%	58	64	110%	102	98	96%
2020-21	43	34	79%	56	64	114%	99	98	99%
2021-22	41	32	77%	53	63	119%	94	95	101%
2022-23	42	25	60%	51	51	100%	93	77	82%
2023-24	41	32	78%	50	63	126%	91	95	104%

Notes:

1. *High-reliability water share volumes prior to 2009-10 are volumes as reported from the 2018 trends update. Water use volumes for 2004-05 are as reported from the 2018 trends update.*

Table 9: Summary of private annual allocation trade in northern Victoria

Water year	Volume traded (GL)			Number of trades			Percentage (%)	Average parcel size (ML)		
	Commercial trades	$0 priced trades	**Total volume**	Commercial trades	$0 priced trades	**Total trades**	Total volume as proportion of private available allocation	Commercial trades	$0 priced trades	All trades
2007-08	371	168	**539**	12,996	4,707	**17,703**	38%	29	36	30
2008-09	460	159	**618**	10,810	4,484	**15,294**	59%	43	35	40
2009-10	448	294	**741**	7,277	3,080	**10,357**	39%	62	95	72
2010-11	404	372	**776**	2,209	1,657	**3,866**	28%	183	225	201
2011-12	715	436	**1,150**	3,513	2,283	**5,796**	28%	204	191	198
2012-13	1,025	369	**1,394**	7,955	3,832	**11,787**	40%	129	96	118
2013-14	902	382	**1,284**	8,410	4,022	**12,432**	58%	107	95	103
2014-15	855	455	**1,310**	9,446	4,653	**14,099**	56%	91	98	93
2015-16	794	523	**1,317**	9,424	5,133	**14,557**	66%	84	102	90
2016-17	928	766	**1,694**	7,571	4,517	**12,088**	83%	123	170	140
2017-18	1,088	820	**1,908**	10,454	5,051	**15,505**	76%	104	162	123
2018-19	1,038	716	**1,754**	14,571	5,194	**19,765**	82%	71	138	89
2019-20	914	965	**1,878**	12,518	7,175	**19,693**	119%	73	134	95
2020-21	1,212	884	**2,096**	11,151	4,667	**15,818**	91%	109	189	133
2021-22	1,224	820	**2,044**	9,510	3,802	**13,312**	81%	129	216	154
2022-23	1,252	911	**2,163**	7,039	2,856	**9,895**	88%	178	319	219
2023-24	1,445	1,009	**2,454**	10,162	3,614	**13,776**	81%	142	279	178

Notes:

1. *The trade values reported in this table exclude environmental trades.*

Table 10: Northern private water availability and use

Water Year	Allocation carried over (GL)	New seasonal allocations (GL)	Net trade in (GL)	Use (GL)
2007-08	121	1,286	11	1,001
2008-09	247	798	192	925
2009-10	246	1,676	62	1,135
2010-11	766	2,053	242	751
2011-12	2,126	1,921	51	1,732
2012-13	1,735	1,750	28	2,402
2013-14	513	1,713	359	1,857
2014-15	622	1,703	183	2,033
2015-16	413	1,592	337	1,762
2016-17	350	1,698	376	1,454
2017-18	840	1,675	-12	1,967
2018-19	472	1,664	56	1,751
2019-20	373	1,209	215	1,112
2020-21	623	1,672	288	1,545
2021-22	501	2,014	229	1,564
2022-23	13	2,448	226	1,272
2023-24	686	2,350	57	1,716

Notes:

1. *'Allocation carried over' reflects the carried over water available after deductions from spillable water accounts.*
2. *There was no available allocation carried over in 2022-23 as 100% of carryover water spilled in accordance with the carryover rules.*

Table 11: Goulburn and Murray private water availability, use, & allocation price

Water year	Allocation carried over (GL)	New seasonal allocations (GL)	Net trade in (GL)	Use (GL)	Median allocation price ($ / ML)
2007-08	121	1,217	11	977	320
2008-09	238	765	190	907	320
2009-10	243	1,644	60	1,126	160
2010-11	760	1,952	250	744	26
2011-12	2,093	1,848	45	1,705	20
2012-13	1,706	1,680	36	2,366	45
2013-14	490	1,643	369	1,825	75
2014-15	603	1,633	193	1,993	122
2015-16	400	1,536	318	1,708	230
2016-17	343	1,618	356	1,398	48
2017-18	819	1,601	8	1,932	125
2018-19	458	1,599	70	1,715	430
2019-20	363	1,154	220	1,084	580
2020-21	611	1,596	275	1,494	100
2021-22	480	1,943	206	1,510	65
2022-23	0	2,367	237	1,255	20
2023-24	669	2,269	70	1,688	40

Notes:

1. *Only privately held accounts with a Goulburn or Murray source trading zone are included in this data. (Trading zones 1A, 1B, 3, 6, 6B or 7.)*
2. *'Allocation carried over' reflects the carried over water available after deductions from spillable water accounts.*
3. *Median allocation price reflects only private priced trades in zones 1A, 1B, 3, 6, 6B, and 7.*
4. *There was no available allocation carried over in 2022-23 as 100% of carryover water spilled in accordance with the carryover rules.*

Table 12: Net allocation trade into Victoria (excluding within environment trade)

Water year	Net volume traded in (GL)		
	From NSW	From SA	Total from Interstate
2007-08	57	-45	11
2008-09	230	-18	212
2009-10	93	-27	66
2010-11	38	207	245
2011-12	-54	46	-8
2012-13	-141	-4	-145
2013-14	120	74	194
2014-15	5	61	66
2015-16	197	51	248
2016-17	119	125	244
2017-18	-146	-7	-153
2018-19	-62	-29	-91
2019-20	106	21	128
2020-21	195	-4	192
2021-22	133	-44	89
2022-23	13	24	37
2023-24	-14	-87	-101

Notes:

1. *The presented volumes exclude 'Within environment' trades i.e. trades between the Commonwealth, VEWH and MDBA accounts and interstate trades with a $0 price involving environmental party on the Victorian side of the trade.*
2. *Negative values in this table indicate trade out of Victoria to interstate destinations.*
3. *Net trade in this figure includes interstate tagged usage.*

Table 13: Net allocation trade into private Murray zone 7 accounts (including tagged use)

Water year	VIC Murray - Dart to Barmah (GL)	Lower Broken Creek (GL)	Victorian Tributaries (GL)	NSW above the Barmah Narrows (GL)	NSW below the Barmah Narrows (GL)	NSW Murrumbidgee (GL)	SA Murray (GL)
2008-09	5	1	29	7	44	94	-5
2009-10	4	1	8	12	32	16	-8
2010-11	1	1	30	-2	13	4	205
2011-12	1	0	-9	-6	-31	-11	36
2012-13	33	5	48	-27	-41	-25	6
2013-14	4	1	48	20	36	19	51
2014-15	12	1	111	-2	6	1	52
2015-16	14	3	57	41	55	39	36
2016-17	-1	0	1	4	39	36	94
2017-18	36	4	211	-23	-35	-53	-2
2018-19	51	5	209	-31	-20	12	-21
2019-20	46	4	155	-3	95	8	1
2020-21	14	-1	17	0	168	31	-11
2021-22	28	8	51	-1	100	41	-46
2022-23	46	6	-2	0	12	0	11
2023-24	38	11	153	0	-15	18	-85

Notes:

1. *Victorian tributaries include trading zones* 1A, 1B, 2, 3, 4A, 4C, and 5A. NSW upstream Barmah includes zones 10 and 10B. NSW downstream Barmah includes zones 11 and 14.

Table 14: June net allocation trade into private Murray zone 7 accounts (including tagged use)

Year	Vic Murray above Barmah	Lower Broken Creek	Vic Tributaries	NSW Murray upstream Barmah	NSW Murray downstream Barmah	NSW Murrumbidgee	SA Murray	Net trade in
June 2009	0	0	1	0	7	5	-1	13
June 2010	0	0	-3	0	0	0	-2	-5
June 2011	-3	0	-1	0	0	0	146	142
June 2012	-2	0	-5	0	0	0	40	34
June 2013	3	1	7	0	0	0	2	12
June 2014	-3	0	0	0	0	0	4	1
June 2015	-2	0	-1	0	0	0	24	21
June 2016	-3	0	-4	0	0	0	-2	-9
June 2017	-6	-1	-47	0	1	0	8	-44
June 2018	-3	0	-4	-2	0	0	-11	-21
June 2019	-11	0	-9	8	0	0	-5	-18
June 2020	-21	-3	-35	-3	-4	0	36	-30
June 2021	-18	-2	-38	1	8	0	27	-23
June 2022	21	0	-41	-1	1	0	6	-14
June 2023	-19	0	-49	0	0	0	13	-55
June 2024	0	-1	-22	0	0	0	2	-22

Notes:

1. *Victorian tributaries include trading zones 1A, 1B, 2, 3, 4A, 4C, and 5A. NSW upstream Barmah includes zones 10 and 10B. NSW downstream Barmah includes zones 11 and 14.*
2. *The large volume of water transferred from SA Murray in 2011 was mainly due to trade from the SA Department for Water to private users. Figure 9 in the body of the report only shows data from June 2016 to make recent trends more visible in light of the large 2011 volumes.*

Table 15: Commercial allocation trade for carryover parking compared to other reasons for trade

Water year	Carryover parking (GL)	Other commercial (GL)	No data available (GL)	Carryover parking median price ($/ML)	Overall median price ($/ML)
2020-21	114	976	122	45	105
2021-22	128	1,059	37	50	70
2022-23	224	1,003	25	65	20
2023-24	167	1,227	51	70	40

Notes:

1. *Data excludes all within and into environment trades.*
2. *2020-21 was the year collection for 'reason for trade' data commenced.*
3. *Median allocation price reflects only private priced trades in zones 1A, 1B, 3, 6, 6B, and 7.*

Table 16: How allocations made to private water shares not linked to land were used

Water year	Sold commercially	Moved to another account for $0	Lost to write-off/evaporation	Carried over to the following year	
				Allocation vol (GL)	% of available water
2007-08	45	21	17	37	30%
2008-09	39	21	10	33	32%
2009-10	26	34	1	29	33%
2010-11	35	37	3	62	45%
2011-12	85	35	13	48	27%
2012-13	87	49	7	21	13%
2013-14	90	46	4	9	6%
2014-15	149	76	3	12	5%
2015-16	148	83	5	51	18%
2016-17	226	165	6	110	22%
2017-18	293	238	5	59	10%
2018-19	351	210	6	65	10%
2019-20	222	266	8	121	20%
2020-21	311	240	13	171	23%
2021-22	362	285	19	216	24%
2022-23	396	349	24	317	29%
2023-24	538	459	30	270	21%

Notes:

1. *The 'available water' in this data includes new season allocations and allocation carried over from the previous year adjusted for spills.*
2. *'Sold commercially' reflects the net volume of commercial trades out of privately held accounts with water shares not linked to land.*
3. *"Moved to another account for $0' reflects the net volume of $0 priced allocation trades out of privately held accounts not linked to land.*
4. *This data differs from equivalent reporting in the 2018 Market Trends Report which was based on water shares now tied to land, rather than not linked to land. Further analysis in how these reporting methods differ is presented in Appendix 1.*

Table 17: How allocations made to private water users were carried over

Water year	Carried over to the following year	
	Allocation vol (GL)	% of available water
2007-08	238	15%
2008-09	224	16%
2009-10	691	32%
2010-11	1,844	66%
2011-12	1,803	41%
2012-13	883	22%
2013-14	592	20%
2014-15	379	13%
2015-16	431	16%
2016-17	689	26%
2017-18	385	13%
2018-19	282	11%
2019-20	455	21%
2020-21	684	24%
2021-22	735	25%
2022-23	849	31%
2023-24	853	26%

Notes:

1. *The 'available water' in this data includes new season allocations and allocation carried over from the previous year adjusted for spills.*

Table 18: Private water users - where they source water from and how that compares to use

Water year	Allocation carried over (GL)	New seasonal allocations (GL)	Bought commercially (net GL)	Moved from another account for $0 (net GL)	Use (GL)	Median allocation price ($)
2007-08	108	1,131	43	30	1,001	320
2008-09	218	694	223	31	925	320
2009-10	215	1,459	108	59	1,135	160
2010-11	618	1,680	145	172	751	26
2011-12	1,878	1,656	267	-38	1,732	20
2012-13	1,550	1,515	231	75	2,402	45
2013-14	461	1,463	493	88	1,857	75
2014-15	566	1,447	375	63	2,033	122
2015-16	371	1,341	380	138	1,762	230
2016-17	276	1,373	453	103	1,454	48
2017-18	686	1,362	312	31	1,967	125
2018-19	379	1,298	284	102	1,751	430
2019-20	281	926	357	42	1,112	580
2020-21	452	1,260	495	85	1,545	100
2021-22	369	1,465	484	69	1,564	65
2022-23	4	1,721	449	40	1,272	20
2023-24	467	1,640	528	38	1,716	40

Notes:

1. *Only privately held northern Victorian accounts with water use recorded in the relevant year are included in this data.*
2. *'Allocation carried over' reflects the carried over water available after deductions from spillable water accounts.*
3. *'Bought commercially' reflects the net volume of commercial trades into private water user accounts.*
4. *"Moved from another account for $0' reflects the net volume of $0 priced trades into private water user accounts.*
5. *There was no net trade out in any year.*

Table 19: Where did northern Victorian irrigators source their water from each year?

Water year	Private category	Allocation carried over (GL)	New seasonal allocations (GL)	Net trade in (GL)	Use (GL)
2007-08	GMW Districts	40	797	-43	657
	GMW Diverters	10	102	-14	61
	LMW Districts	21	91	30	82
	LMW Diverters	37	141	98	201
	Subtotal	**108**	**1,131**	**72**	**1,001**
2008-09	GMW Districts	117	473	93	556
	GMW Diverters	19	52	0	43
	LMW Districts	29	52	45	89
	LMW Diverters	53	116	115	237
	Subtotal	**218**	**694**	**254**	**925**
2009-10	GMW Districts	120	1,029	58	745
	GMW Diverters	14	75	-7	33
	LMW Districts	35	109	1	87
	LMW Diverters	46	246	115	269
	Subtotal	**215**	**1,459**	**167**	**1,135**
2010-11	GMW Districts	410	1,187	105	495
	GMW Diverters	31	87	2	19
	LMW Districts	52	133	-1	43
	LMW Diverters	125	273	211	194
	Subtotal	**618**	**1,680**	**317**	**751**
2011-12	GMW Districts	1,201	1,171	199	1,269
	GMW Diverters	104	102	2	58
	LMW Districts	139	128	-7	84
	LMW Diverters	434	255	34	320
	Subtotal	**1,878**	**1,656**	**229**	**1,732**
2012-13	GMW Districts	935	1,048	297	1,777
	GMW Diverters	115	120	-39	104
	LMW Districts	153	122	-49	114
	LMW Diverters	347	224	97	407
	Subtotal	**1,550**	**1,515**	**306**	**2,402**
2013-14	GMW Districts	381	999	431	1,317
	GMW Diverters	44	117	-15	88
	LMW Districts	9	119	-1	98
	LMW Diverters	27	228	166	354
	Subtotal	**461**	**1,463**	**581**	**1,857**
2014-15	GMW Districts	451	981	305	1,451
	GMW Diverters	40	116	-21	89
	LMW Districts	20	118	-15	94
	LMW Diverters	54	232	168	399
	Subtotal	**566**	**1,447**	**437**	**2,033**
2015-16	GMW Districts	264	907	267	1,157
	GMW Diverters	33	104	-14	79
	LMW Districts	26	118	6	102

Water year	Private category	Allocation carried over (GL)	New seasonal allocations (GL)	Net trade in (GL)	Use (GL)
	LMW Diverters	48	212	260	424
	Subtotal	**371**	**1,341**	**519**	**1,762**
2016-17	GMW Districts	238	934	372	929
	GMW Diverters	24	114	-5	64
	LMW Districts	0	118	-3	87
	LMW Diverters	14	207	192	374
	Subtotal	**276**	**1,373**	**556**	**1,454**
2017-18	GMW Districts	572	900	186	1,341
	GMW Diverters	51	115	-24	93
	LMW Districts	16	115	-10	101
	LMW Diverters	46	232	191	432
	Subtotal	**686**	**1,362**	**342**	**1,967**
2018-19	GMW Districts	296	873	133	1,093
	GMW Diverters	37	111	-23	86
	LMW Districts	18	109	6	110
	LMW Diverters	28	205	268	462
	Subtotal	**379**	**1,298**	**385**	**1,751**
2019-20	GMW Districts	192	622	71	531
	GMW Diverters	27	83	-13	51
	LMW Districts	20	70	35	98
	LMW Diverters	42	151	304	433
	Subtotal	**281**	**926**	**398**	**1,112**
2020-21	GMW Districts	332	819	337	942
	GMW Diverters	36	114	-11	69
	LMW Districts	25	101	14	98
	LMW Diverters	59	225	240	436
	Subtotal	**452**	**1,260**	**580**	**1,545**
2021-22	GMW Districts	331	1,006	281	956
	GMW Diverters	31	121	-2	65
	LMW Districts	0	99	19	93
	LMW Diverters	7	239	254	449
	Subtotal	**369**	**1,465**	**553**	**1,564**
2022-23	GMW Districts	-7	1,259	315	816
	GMW Diverters	7	124	5	53
	LMW Districts	0	91	22	75
	LMW Diverters	3	247	147	329
	Subtotal	**4**	**1,721**	**489**	**1,272**
2023-24	GMW Districts	405	1,182	297	1,130
	GMW Diverters	35	130	-3	73
	LMW Districts	7	91	32	94
	LMW Diverters	20	237	240	418
	Subtotal	**467**	**1,640**	**566**	**1,716**

Notes:

1. Only privately held accounts with water use recorded against them during the relevant year are included in this data.

Table 20: Water sourced by irrigators as a percentage of water use

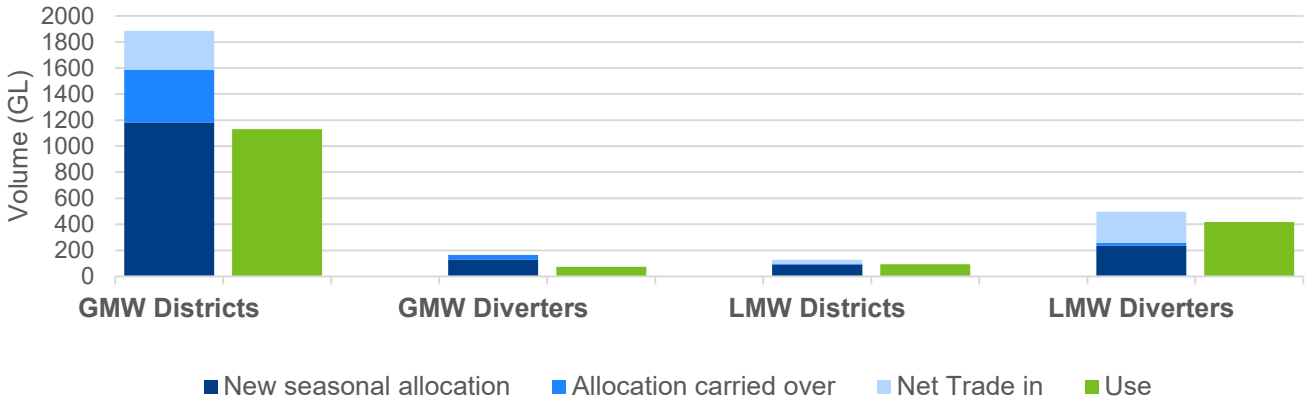
Water year	Private category	Carried over from previous year	Seasonal allocation	Net trade in	Use (GL)
2007-08	GMW Districts	6%	121%	-7%	657
	GMW Diverters	16%	167%	-23%	61
	LMW Districts	26%	111%	37%	82
	LMW Diverters	18%	70%	49%	201
	Subtotal	**11%**	**113%**	**7%**	**1,001**
2008-09	GMW Districts	21%	85%	17%	556
	GMW Diverters	45%	122%	0%	43
	LMW Districts	33%	58%	50%	89
	LMW Diverters	22%	49%	49%	237
	Subtotal	**24%**	**75%**	**27%**	**925**
2009-10	GMW Districts	16%	138%	8%	745
	GMW Diverters	42%	226%	-20%	33
	LMW Districts	40%	125%	1%	87
	LMW Diverters	17%	92%	43%	269
	Subtotal	**19%**	**129%**	**15%**	**1,135**
2010-11	GMW Districts	83%	240%	21%	495
	GMW Diverters	163%	459%	11%	19
	LMW Districts	122%	311%	-3%	43
	LMW Diverters	64%	141%	109%	194
	Subtotal	**82%**	**224%**	**42%**	**751**
2011-12	GMW Districts	95%	92%	16%	1,269
	GMW Diverters	180%	176%	4%	58
	LMW Districts	166%	153%	-8%	84
	LMW Diverters	135%	80%	11%	320
	Subtotal	**108%**	**96%**	**13%**	**1,732**
2012-13	GMW Districts	53%	59%	17%	1,777
	GMW Diverters	110%	116%	-38%	104
	LMW Districts	135%	107%	-43%	114
	LMW Diverters	85%	55%	24%	407
	Subtotal	**65%**	**63%**	**13%**	**2,402**
2013-14	GMW Districts	29%	76%	33%	1,317
	GMW Diverters	50%	133%	-16%	88
	LMW Districts	9%	121%	-1%	98
	LMW Diverters	8%	64%	47%	354
	Subtotal	**25%**	**79%**	**31%**	**1,857**
2014-15	GMW Districts	31%	68%	21%	1,451
	GMW Diverters	45%	129%	-23%	89
	LMW Districts	21%	126%	-16%	94
	LMW Diverters	13%	58%	42%	399
	Subtotal	**28%**	**71%**	**22%**	**2,033**
2015-16	GMW Districts	23%	78%	23%	1,157
	GMW Diverters	42%	132%	-18%	79
	LMW Districts	25%	116%	6%	102

Water year	Private category	Carried over from previous year	Seasonal allocation	Net trade in	Use (GL)
	LMW Diverters	11%	50%	61%	424
	Subtotal	**21%**	**76%**	**29%**	**1,762**
2016-17	GMW Districts	26%	100%	40%	929
	GMW Diverters	37%	177%	-8%	64
	LMW Districts	0%	136%	-3%	87
	LMW Diverters	4%	55%	51%	374
	Subtotal	**19%**	**94%**	**38%**	**1,454**
2017-18	GMW Districts	43%	67%	14%	1,341
	GMW Diverters	55%	124%	-26%	93
	LMW Districts	16%	114%	-10%	101
	LMW Diverters	11%	54%	44%	432
	Subtotal	**35%**	**69%**	**17%**	**1,967**
2018-19	GMW Districts	27%	80%	12%	1,093
	GMW Diverters	43%	129%	-27%	86
	LMW Districts	16%	98%	6%	110
	LMW Diverters	6%	44%	58%	462
	Subtotal	**22%**	**74%**	**22%**	**1,751**
2019-20	GMW Districts	36%	117%	13%	531
	GMW Diverters	54%	163%	-25%	51
	LMW Districts	20%	71%	36%	98
	LMW Diverters	10%	35%	70%	433
	Subtotal	**25%**	**83%**	**36%**	**1,112**
2020-21	GMW Districts	35%	87%	36%	942
	GMW Diverters	53%	166%	-15%	69
	LMW Districts	25%	104%	14%	98
	LMW Diverters	14%	52%	55%	436
	Subtotal	**29%**	**82%**	**38%**	**1,545**
2021-22	GMW Districts	35%	105%	29%	956
	GMW Diverters	47%	185%	-3%	65
	LMW Districts	0%	106%	21%	93
	LMW Diverters	2%	53%	57%	449
	Subtotal	**24%**	**94%**	**35%**	**1,564**
2022-23	GMW Districts	-1%	154%	39%	816
	GMW Diverters	14%	233%	9%	53
	LMW Districts	0%	122%	30%	75
	LMW Diverters	1%	75%	45%	329
	Subtotal	**0%**	**135%**	**38%**	**1,272**
2023-24	GMW Districts	36%	105%	26%	1,130
	GMW Diverters	47%	179%	-4%	73
	LMW Districts	7%	96%	33%	94
	LMW Diverters	5%	57%	57%	418
	Subtotal	**27%**	**96%**	**33%**	**1,716**

Notes:

1. *Only privately held accounts with water use recorded against them during the relevant year are included in this data.*

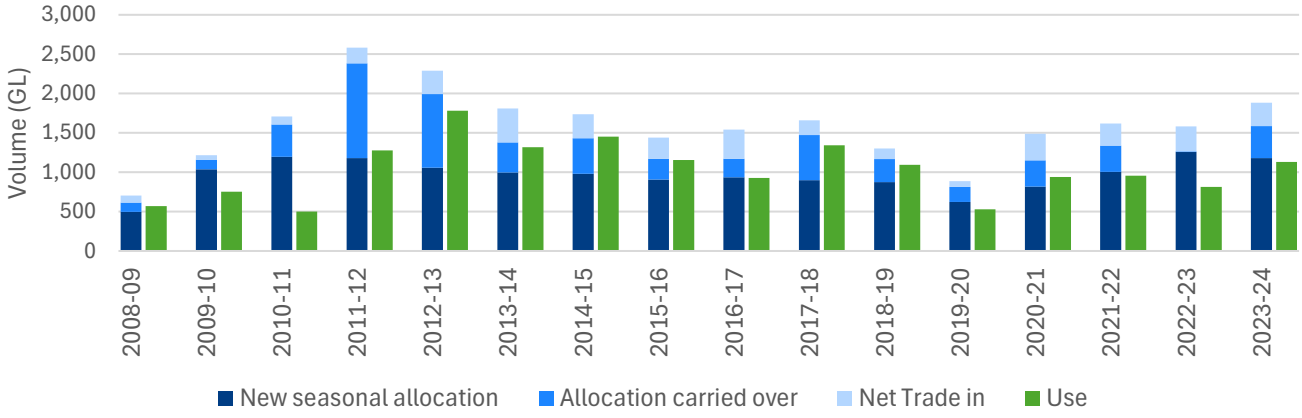
Figure 21: Where did northern Victorian irrigators source their water from in 2023-24



Notes:

1. *Only privately held accounts with water use recorded against them during the relevant year are included in this data.*

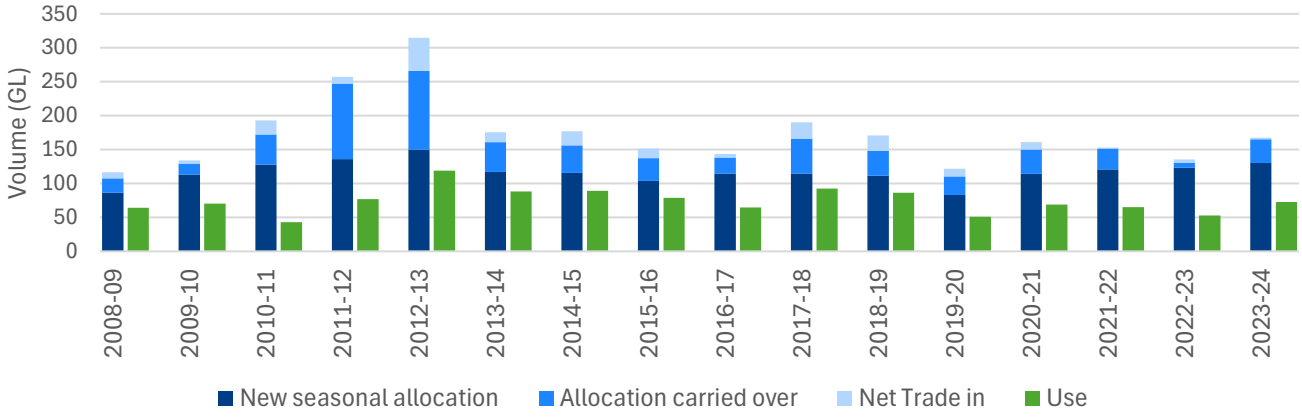
Figure 22: GMW district irrigators source of allocation



Notes:

1. *Only privately held accounts with water use recorded against them during the relevant year are included in this data.*

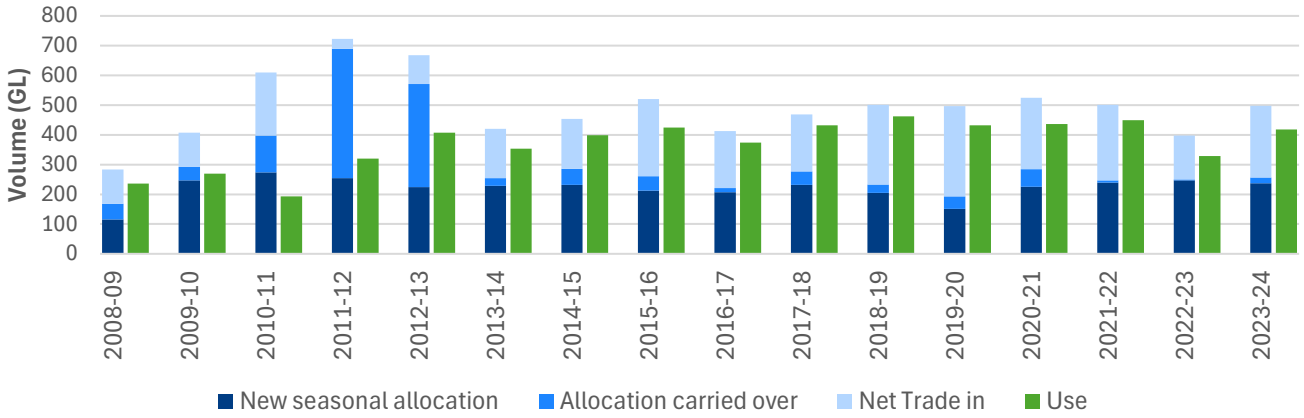
Figure 23: GMW diverter irrigators source of allocation



Notes:

1. *Only privately held accounts with water use recorded against them during the relevant year are included in this data.*

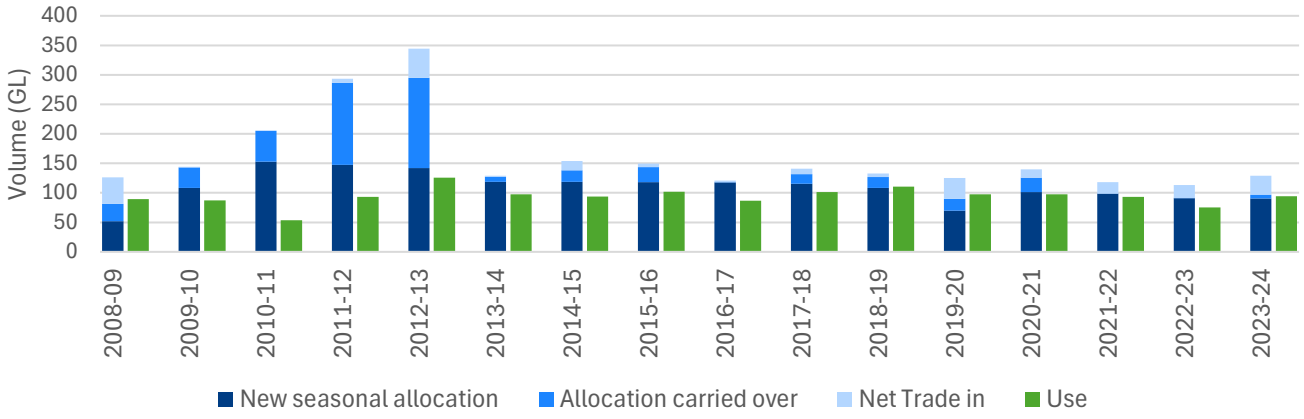
Figure 24: LMW diverter irrigators source of allocation



Notes:

1. *Only privately held accounts with water use recorded against them during the relevant year are included in this data.*

Figure 25: LMW district irrigators source of allocation



Notes:

1. *Only privately held accounts with water use recorded against them during the relevant year are included in this data.*

Table 21: Commercial allocation trade in Northern Victoria categorised by buyer and seller type

Trade to (buyer):	Private					Interstate				Environment				Water Corporation					Grand Total
Trade from (seller):	Private	Interstate	Water Corp	Environment	**Total**	Private	Water Corp	Environment	**Total**	Private	Interstate	Water Corp	**Total**	Private	Interstate	Water Corp	Environment	**Total**	
2007-2008	234	60	8	-	**302**	57	-	-	**57**	-	-	-	**0**	8	3	-	-	**12**	**371**
2008-2009	186	214	8	-	**407**	24	-	-	**24**	-	-	-	**0**	12	15	3	-	**29**	**460**
2009-2010	274	95	9	-	**378**	55	-	-	**55**	-	-	-	**0**	8	3	3	-	**14**	**448**
2010-2011	213	108	26	-	**346**	25	1	-	**26**	8	20	-	**28**	3	-	0	-	**3**	**404**
2011-2012	512	114	43	10	**678**	34	2	-	**36**	-	-	-	**0**	0	-	-	-	**0**	**715**
2012-2013	652	63	129	7	**850**	140	28	6	**174**	-	-	-	**0**	0	-	-	-	**0**	**1,025**
2013-2014	522	200	154	-	**876**	18	7	-	**26**	0	-	-	**0**	1	-	0	-	**1**	**902**
2014-2015	557	115	90	9	**771**	47	22	3	**72**	-	10	-	**10**	2	-	-	-	**2**	**855**
2015-2016	507	168	59	22	**757**	27	10	-	**37**	-	-	-	**0**	0	-	-	-	**0**	**794**
2016-2017	624	142	116	20	**901**	20	5	0	**26**	-	-	-	**0**	2	-	-	-	**2**	**928**
2017-2018	783	43	105	10	**941**	126	11	5	**142**	-	-	-	**0**	4	-	-	-	**4**	**1,088**
2018-2019	750	68	77	30	**924**	106	8	1	**114**	-	-	-	**0**	0	-	-	-	**0**	**1,038**
2019-2020	581	192	79	-	**852**	54	7	-	**62**	-	-	-	**0**	-	-	-	-	**0**	**914**
2020-2021	822	230	83	-	**1,135**	64	5	-	**69**	-	-	-	**0**	8	-	-	-	**8**	**1,212**
2021-2022	840	143	134	11	**1,128**	74	2	1	**76**	-	-	-	**0**	19	0	0	-	**19**	**1,224**
2022-2023	934	44	166	44	**1,188**	34	3	1	**38**	1	-	-	**1**	25	-	0	-	**26**	**1,252**
2023-2024	1,117	57	122	28	**1,323**	77	17	8	**103**	6	-	-	**6**	13	-	-	-	**13**	**1,445**

Notes:

1. *Only priced allocation trades (where the recorded price was greater than $0) are included in this table.*
2. *Values for 2007-08 are as reported in the 2018 trends report.*

Table 22: Percentage of commercial allocation trade in Northern Victoria categorised by buyer and seller type

Trade to (buyer):	Private					Interstate				Environment				Water Corporation					Grand Total
Trade from (seller):	Private	Interstate	Water Corp	Environment	**Total**	Private	Water Corp	Environment	**Total**	Private	Interstate	Water Corp	**Total**	Private	Interstate	Water Corp	Environment	**Total**	
2007-2008	63%	16%	2%	-	**81%**	15%	-	-	**15%**	-	-	-	**0%**	2%	1%	-	-	**3%**	**100%**
2008-2009	40%	47%	2%	-	**88%**	5%	-	-	**5%**	-	-	-	**0%**	3%	3%	1%	-	**6%**	**100%**
2009-2010	61%	21%	2%	-	**84%**	12%	-	-	**12%**	-	-	-	**0%**	2%	1%	1%	-	**3%**	**100%**
2010-2011	53%	27%	6%	-	**86%**	6%	0%	-	**6%**	2%	5%	-	**7%**	1%	-	0%	-	**1%**	**100%**
2011-2012	72%	16%	6%	1%	**95%**	5%	0%	-	**5%**	-	-	-	**0%**	0%	-	-	-	**0%**	**100%**
2012-2013	64%	6%	13%	1%	**83%**	14%	3%	1%	**17%**	-	-	-	**0%**	0%	-	-	-	**0%**	**100%**
2013-2014	58%	22%	17%	-	**97%**	2%	1%	-	**3%**	0%	-	-	**0%**	0%	-	0%	-	**0%**	**100%**
2014-2015	65%	13%	11%	1%	**90%**	5%	3%	0%	**8%**	-	1%	-	**1%**	0%	-	-	-	**0%**	**100%**
2015-2016	64%	21%	7%	3%	**95%**	3%	1%	-	**5%**	-	-	-	**0%**	0%	-	-	-	**0%**	**100%**
2016-2017	67%	15%	13%	2%	**97%**	2%	1%	0%	**3%**	-	-	-	**0%**	0%	-	-	-	**0%**	**100%**
2017-2018	72%	4%	10%	1%	**86%**	12%	1%	0%	**13%**	-	-	-	**0%**	0%	-	-	-	**0%**	**100%**
2018-2019	72%	7%	7%	3%	**89%**	10%	1%	0%	**11%**	-	-	-	**0%**	0%	-	-	-	**0%**	**100%**
2019-2020	64%	21%	9%	-	**93%**	6%	1%	-	**7%**	-	-	-	**0%**	-	-	-	-	**0%**	**100%**
2020-2021	68%	19%	7%	-	**94%**	5%	0%	-	**6%**	-	-	-	**0%**	1%	-	-	-	**1%**	**100%**
2021-2022	69%	12%	11%	1%	**92%**	6%	0%	0%	**6%**	-	-	-	**0%**	2%	0%	0%	-	**2%**	**100%**
2022-2023	75%	4%	13%	4%	**95%**	3%	0%	0%	**3%**	0%	-	-	**0%**	2%	-	0%	-	**2%**	**100%**
2023-2024	77%	4%	8%	2%	**92%**	5%	1%	1%	**7%**	0%	-	-	**0%**	1%	-	-	-	**1%**	**100%**

Notes:

1. *Only priced allocation trades (where the recorded price was greater than $0) are included in this table.*
2. *Values for 2007-08 are as reported in the 2018 trends report.*

Table 23: Proportion of commercial Goulburn and Murray trade involving at least one frequent trader account

Water year	# trades			Volume (GL)		
	Trade involved at least one frequent trader account	All other commercial trades	% involving frequent trader account	Volume of trade involving at least one frequent trader account	All other commercial trades	% involving frequent trader account
2015-16	3,107	6,255	33%	405	387	51%
2016-17	2,645	4,899	35%	441	485	48%
2017-18	4,319	6,065	42%	660	424	61%
2018-19	7,180	7,309	50%	670	364	65%
2019-20	5,769	6,689	46%	592	319	65%
2020-21	3,930	7,183	35%	579	629	48%
2021-22	3,486	5,980	37%	636	586	52%
2022-23	2,615	4,405	37%	726	634	53%
2023-24	3,885	6,225	38%	860	640	57%

Notes:

1. *Frequent trader allocation accounts are identified as individual accounts that have, during a water year, completed more than 20 allocation trades which are: a) approved, b) priced (i.e. commercial), and c) within, from, or to the Goulburn water system (zones 1A, 1B, and 3) or the Murray water system (zones 6, 6B, and 7).*